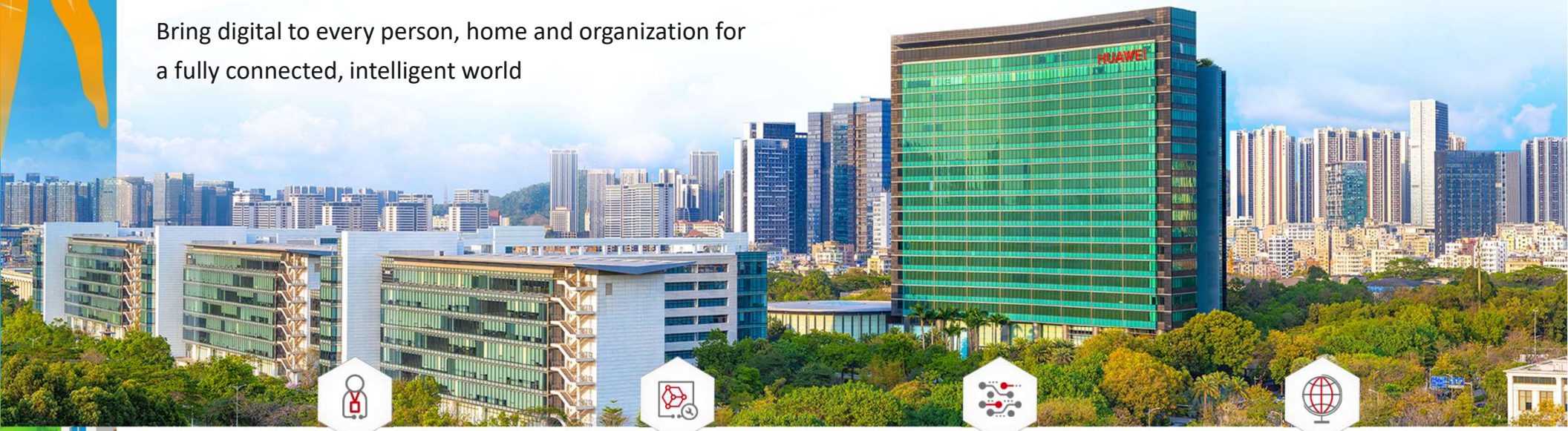
Máster de Formación Permanente
Máster en Energías Renovables y Medio Ambiente
20 EDICIÓN 2025-2026
POLITÉCNICA
UNIVERSIDAD POLITÉCNICA DE MADRID
escuela técnica superior de ingeniería diseño industrial
POLITÉCNICA Máster ERMA
Módulo 5
Sistemas autónomos y Microrredes
La digitalización y la electrónica de potencia en sistemas de almacenamiento energético
Andrés Hernando Ros
HUAWEI

1. **Huawei Digital Power: Updates & Trends**
2. Digitalization as an improvement of LCOS
3. Grid forming
4. Intelligent O&M

Huawei: Leading provider of ICT infrastructure and smart devices

Vision and mission

Bring digital to every person, home and organization for a fully connected, intelligent world



213,000
employees

53.7%
of employees work in R&D

63.8%
of employees outside China hired locally

170+
countries and regions

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Global R&D teams and technology platforms: Leveraging the domain specific advantages globally to keep leading

170
countries/regions

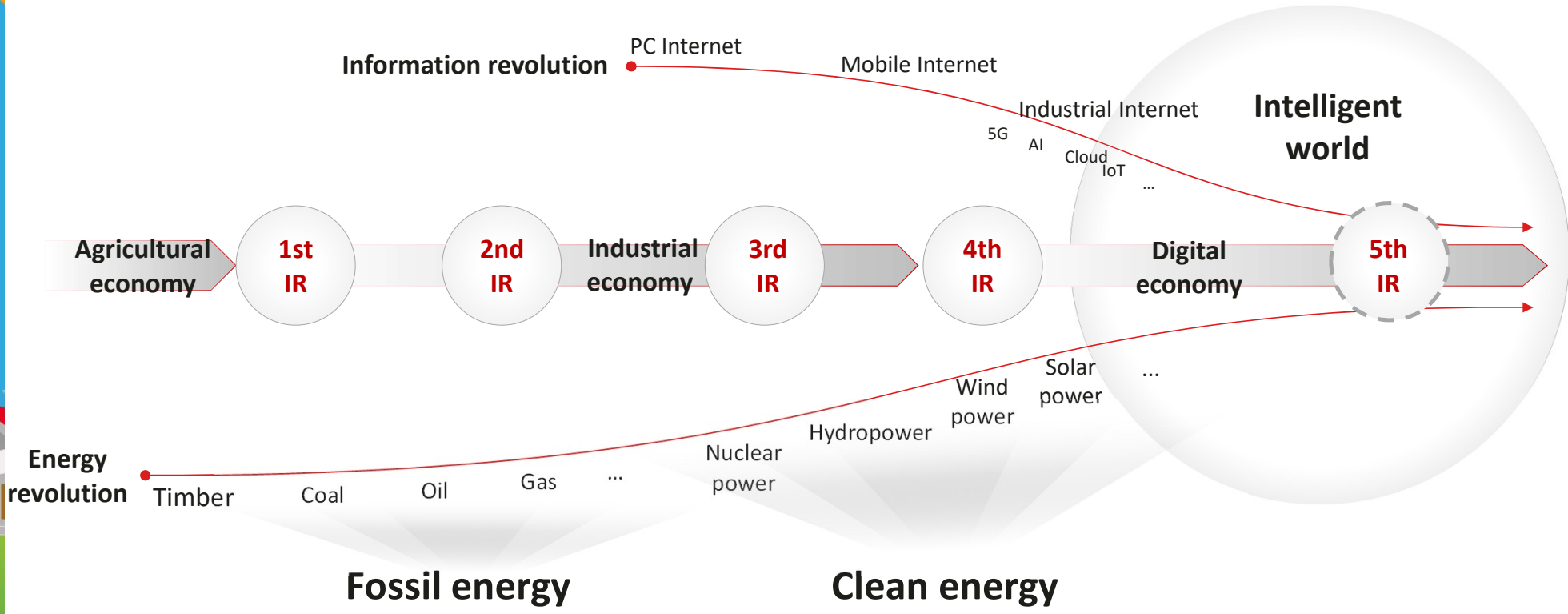
10,000+ employees
60% work in R&D

13
R&D Centers

15%+
as % of revenue in 2024

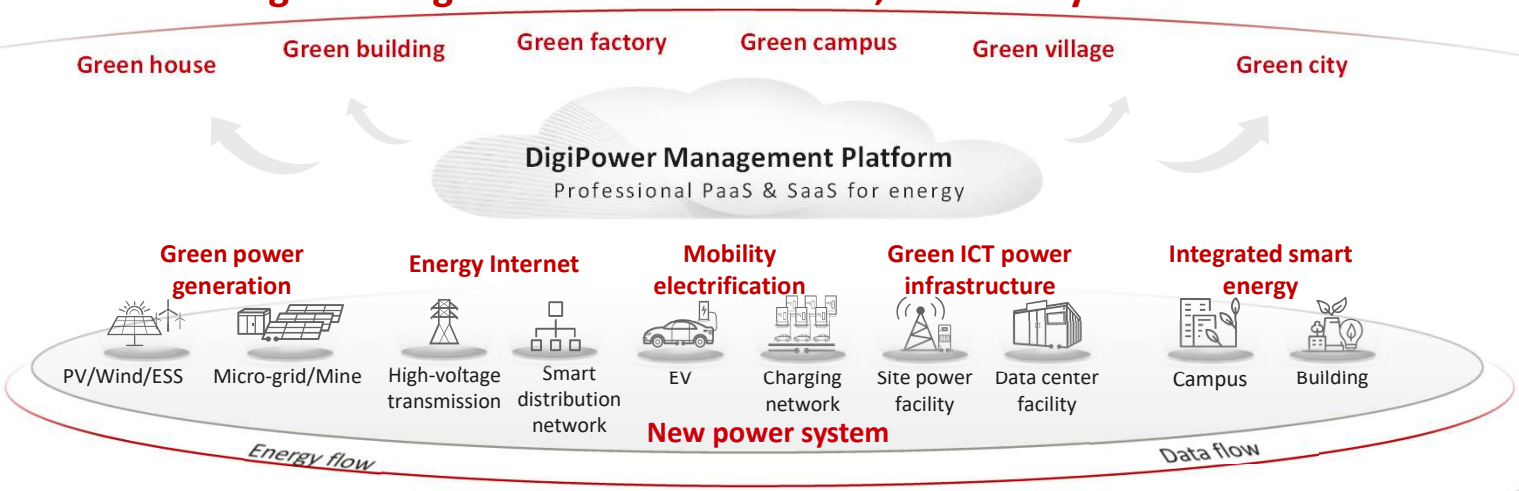
3,484
active patents worldwide

The 5th Industrial Revolution goes hand in hand with low-carbon, digital, and intelligent The 5th Industrial Revolution goes hand in hand with low-carbon, digital, and intelligent transformations, all of which bring us closer to the intelligent world



Huawei Digital Power: Integrating digital and power electronics technologies, developing clean power, and enabling energy digitalization to drive energy revolution for a better, greener future

Evolving from high carbon to low carbon, and finally to net-zero carbon



Focus on products

Power electronics technology (Watt/Heat/Battery)

Material (GaN/SiC) · Component (IGBT/MOS) · Heat management · Energy storage management

Digital technology (Bit)

Sensing (Digital/Visual) · Connection (5G/IoT) · Cloud (Data/Algorithm) · AI (Chip/Algorithm)

Open ecosystem

FusionSolar9.0 Smart String Grid Forming PV&ESS Solution
Start a new era of all-scenario Grid Forming

All-Scenario Grid Forming

ESS grid forming » PV/PV+ESS grid forming

Full-Link Intelligence

Visible & Manageable » AI-enabled planning, construction, maintenance, and operations

High Quality Throughout Lifecycle

Device-level reliability » Full-lifecycle safety and reliability

New

Smart String Grid Forming Inverter
SUN2000-506KTL

New

Smart String Grid Forming ESS
LUNA2000-5015 Series

New

Smart PCS Skid
LUNA2000-PCS-SKID

New

Energy Management System
SPPC+SPMS

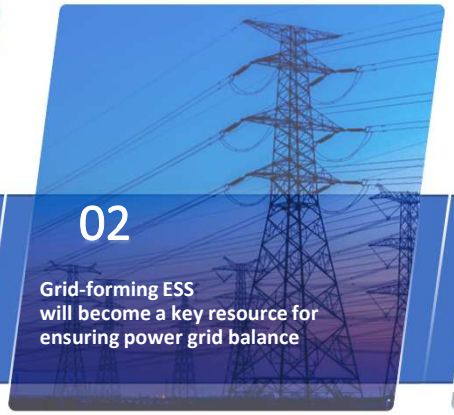
New

PV+ESS AI Agent
FusionSolar Agent

Top 10 Trends of Smart PV & ESS

01

PV+wind+ESS synergy will enable renewables to become predictable, controllable, stable power sources



02

Grid-forming ESS will become a key resource for ensuring power grid balance

03

Generation-grid-load-storage synergy will drive regional autonomy and global coordination of power supply

04

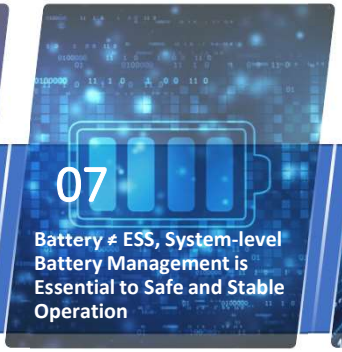
In the Residential PV+ESS scenario, Transition AI-enabled to AI-native, Delivering the Optimal Energy Experience

05

High Frequency and Density Will Drive an Increase in PV+ESS Device Power Density

06

High Voltage and Reliability will Drive LCOE Reduction



07

Battery ≠ ESS, System-level Battery Management is Essential to Safe and Stable Operation

08

A Mature Grid-forming Technology System will Accelerate the Construction of a New Power System

09

AI Agents Will Enable Renewable Power Plant Automation

10

Safety Quantification will Drive Safety Capability Improvement Across the Energy Storage Industry

White Paper on Top 10 Smart PV & ESS Trends

Scan the QR code to download

Challenges in Battery Energy Storage System Industry

Low Available Capacity

- Series & Parallel mismatch due to inconsistency between battery cells, which leads to lower available capacity according to Cannikin Law

Pack & Rack Optimizer

Complex O&M

- On-site battery installation wiring & commissioning
- Regular SOC calibration by professional staff

No need for periodic balancing
No need for experts to visit sites

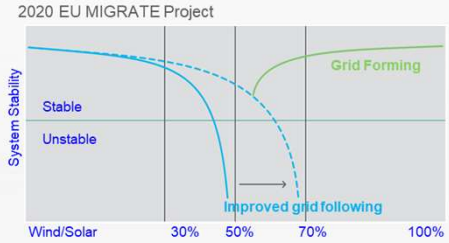
Fire Risks

- Battery cell over-charge, over-discharge ,or other faults
- Key components(circuit boards, contactors, etc.) failure cause sparking and arcing

Cell to system safety protection
Avoid thermal runaway

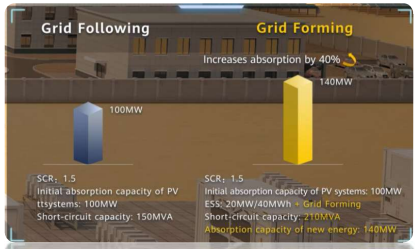
Grid Forming

EU considers grid forming as a key to high wind/solar penetration



The European Power Grid Code (RfG) raised requirements of Grid Forming for Type A-D power generating modules

Grid forming increase absorption of PV capacity by 40%



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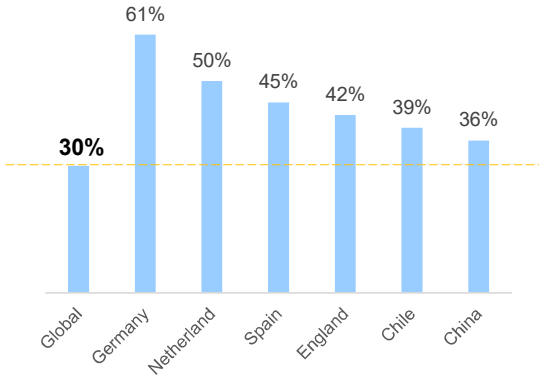
Grid-connection challenges: High penetration renewable energy has led to weak grid and the issue of stability of the power system

Renewable energy penetration rate (PR) exceeds 30% in more and more countries



Typically, **30%** penetration rate is the upper limit of conventional power systems, additional flexible regulating power source is mandatory for higher rate.

High proportion of renewable energy leads to weakening of power grids, serious accidents frequently occurred

(Voltage/ Frequency/ Phase Angle Stability Challenges)

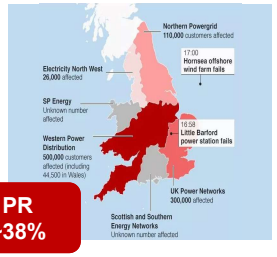
Australia

PR ~50%

South Australia big power outage event last for 50 hours

Voltage dips 6 times

UK

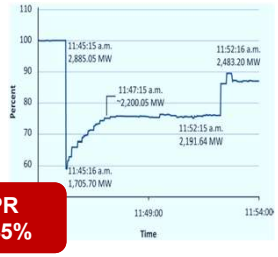


PR ~38%

England and Wales power outages, 1 million people

Frequency drops to 48.9Hz

USA



PR ~45%

Multiple large-scale PV power pla off-grid accidents in California

1200 MW PV being cut-off

Safety challenges: safety hazards become the primary challenge for PV & ESS systems

PV: Bigger power generating higher risks

PV panels and inverters power keeping increasing

MPPT current **< 10A** » MPPT current **> 20A**

Rated Power **< 200kW** ≈ 18 string inputs » Rated Power **> 300kW** ≈ 28 string inputs

PV deployment environment are more and more hazardous

Desert **Mountain** **Agrivoltaics** **Floating** ···

ESS: Safety is the basis of industry development

Main causes of extreme faults in ESS

280Ah » **280、306、314、345、587Ah …**

2000 kWh per 20-ft container » **5000+ kWh** per 20-ft container

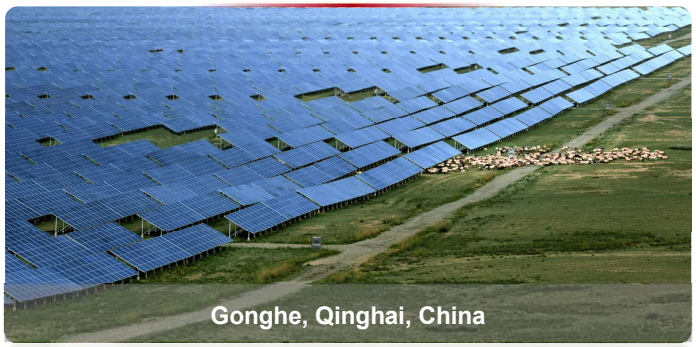
Electrochemical reaction uncertainty

External mechanical abuse

External and internal electrical abuse

O&M challenges: GW-level plant, complex terrains, low intelligence leading to lower system availability and higher O&M cost

SPIC 2.2 GW plant
World's largest string inverter power plant



5 million PV panels, **56** km²

Yalong River 1 GW plant
World's largest and highest-altitude hydro-solar hybrid project

2.1million PV panels, **4600m** altitude, **100km** Farthest PV arrays

Large	Numerous	Harsh	Heavy workload	Long lifetime
10+ million m2 coverage	2+ million PV panels/GW 1+ million cells/GWh	Various terrains, altitude, environment	Lack of intelligence, relying heavily on manual operations	Service time from 20 to 25~35years Spare part difficulty to obtain

Digital @Higher yield **Intelligent** @More output **Automatic** @Less O&M

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1

Foundations of
Battery State Metrics

2

Beyond Estimation:
Accuracy and Consistency
in Battery State Metrics

3

Unlocking Value: Operational Advantages of Accurate SOC

1 Foundations of Battery State Metrics

Why Battery State Metrics Matter

Understanding state metrics is essential for reliable SOH prediction and long-term asset performance

Context: ***Lower LCOS win!***

- Lower CAPEX and lower OPEX
- Higher performance and lifetime expectation
- Better safety and Grid Forming means Less Contingency

Role of State Metrics: ***Common foundation for control, optimization, and maintenance***

- SOC → Energy availability
- SOH → Capacity and degradation
- SOE → Effectively usable energy

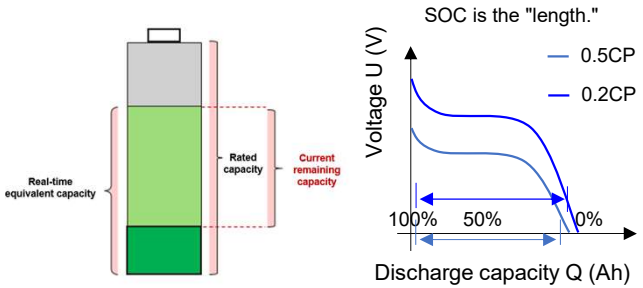
Industry Shift

- From Measurement → To Prediction
- From Simple Estimates → To Advanced Models
- From Reactive Operation → To Data-Driven Operation

SOX Definition and Estimation Formula

SOC

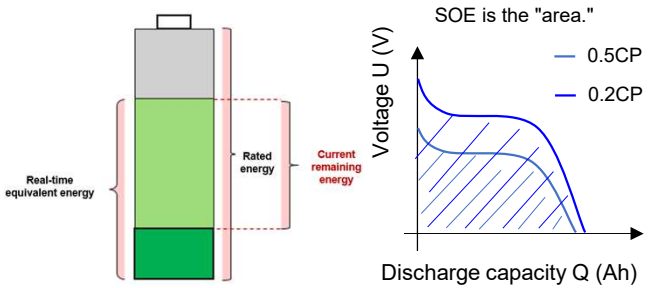
State of charge (SOC) refers to the ratio between the battery's remaining capacity (Ah) and its real-time equivalent capacity (Ah). It is typically expressed as a percentage and indicates the remaining charge in the battery. Since SOC **is not related to the battery voltage** and is minimally affected by the current rate, it is commonly used to control the charge/discharge cutoff for energy storage systems (ESSs), balance between battery racks or packs, and facilitate electricity trading.



$$SOC = \frac{Remainin\ \ capacity\ \ (I \times t)}{Real\text{-}time\ equivalent\ capacity\ \ (I \times t)}$$

SOE

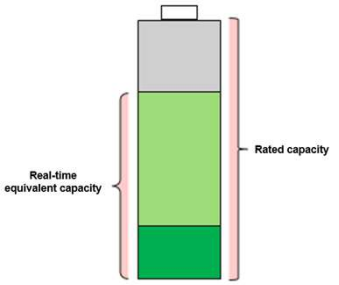
State of energy (SOE) refers to the ratio between the battery's remaining energy (Wh) and its real-time equivalent energy (Wh). It is typically expressed as a percentage and indicates the remaining energy in the battery. **Since SOE is related to the battery voltage** and is greatly affected by the current rate, it is used to estimate the discharge time at the real-time power, similar to the driving range of a car.



$$SOE = \frac{Remainin\ \ energy\ \ (I \times U \times t)}{Real\text{-}time\ equivalent\ energy\ \ (I \times U \times t)}$$

SOH

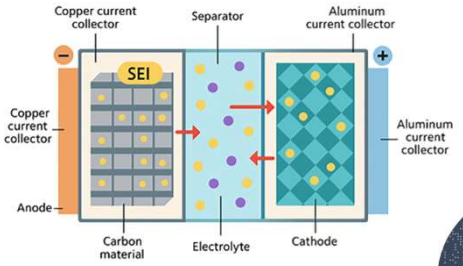
State of health (SOH) refers to the ratio between the battery's current equivalent capacity (Ah) and its rated capacity (Ah). It is typically expressed as a percentage and used to reflect the battery aging degree.

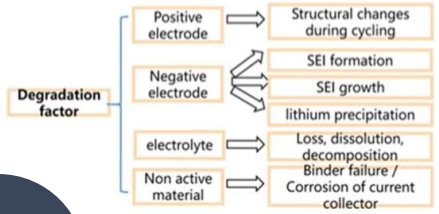


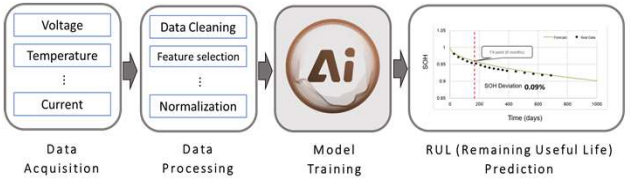
$$SOH = \frac{Real\text{-}time\ equivalent\ capacity\ \ (I \times t)}{Rated\ capacity\ \ (I \times t)}$$

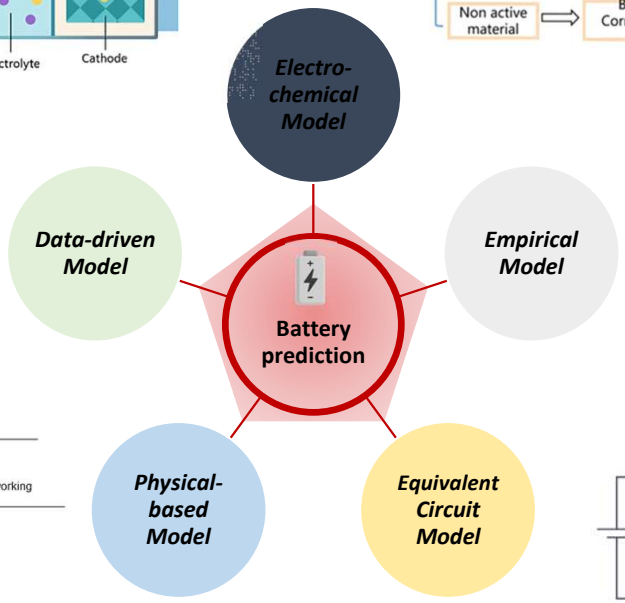
SOH as a strategic compass. Invisible in CAPEX, essential in ROI.

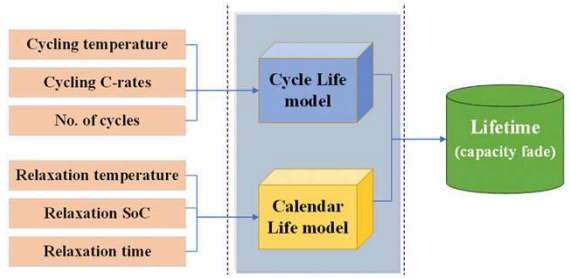
Battery modeling methods to predict the State of Health (SOH)

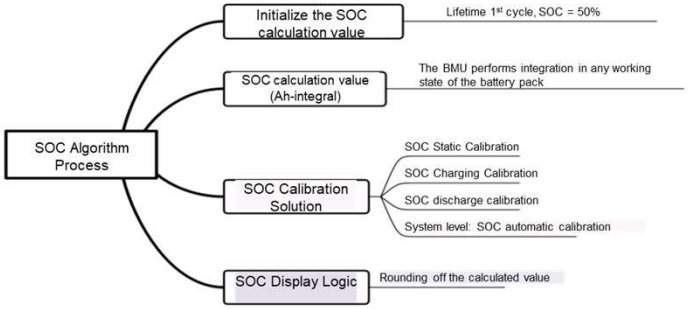




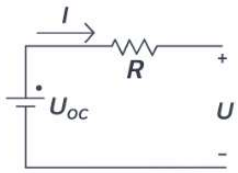


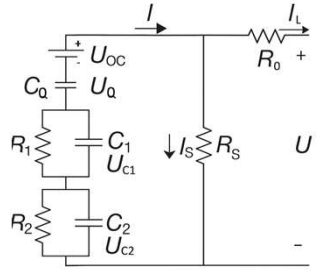






SOC & SOH forecast



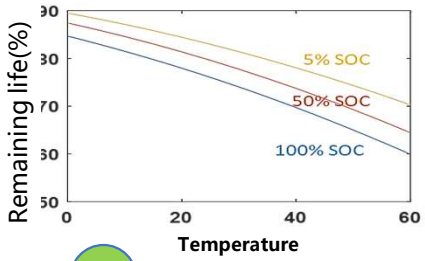


Thevenin, Internal Ohmic Resistance…

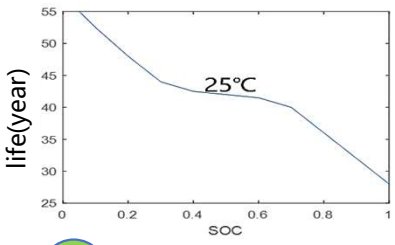
Not all the SOH prediction models are the same

Summary of factors affecting SOH

Calendar Life



1 **Temp. °C**



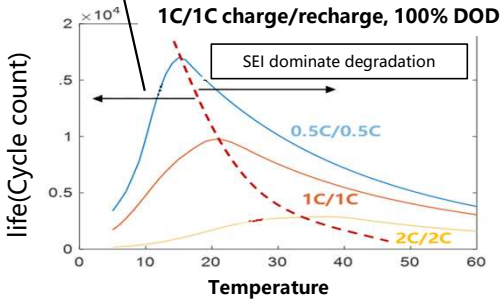
2 **SOC**

Cycle Life

lithium precipitation dominate degradation

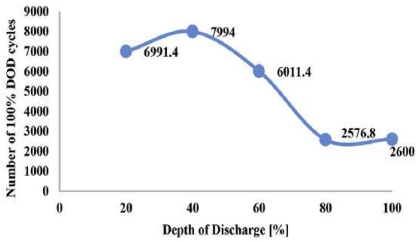
Cycle life vs. temperature

1C/1C charge/recharge, 100% DOD

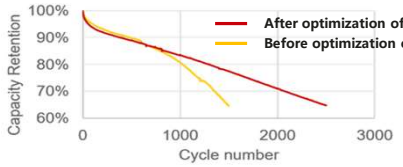


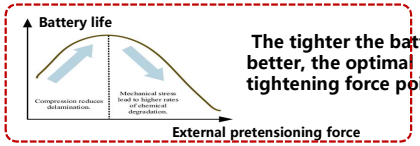
1 **Temp. °C**

2 **C rate**



3 **Rest Interval**



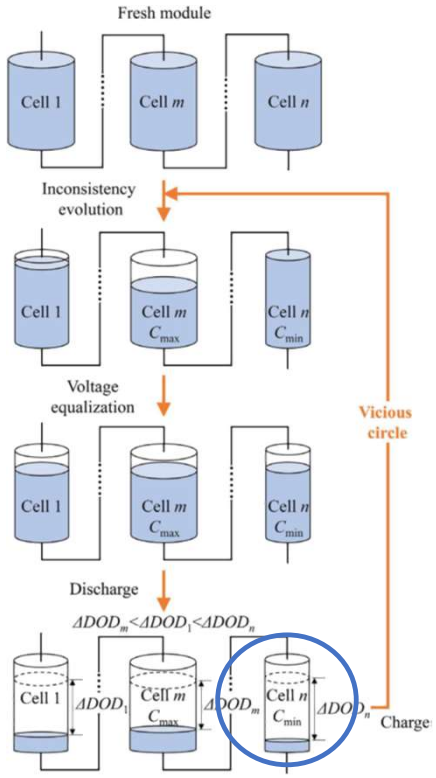


The tighter the battery is, the better, the optimal pre-tightening force point exists!

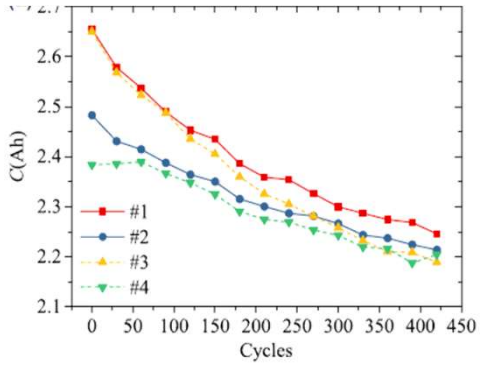
4 **Expansion force**

2 Beyond Estimation: Accuracy and Consistency in Battery State Metrics

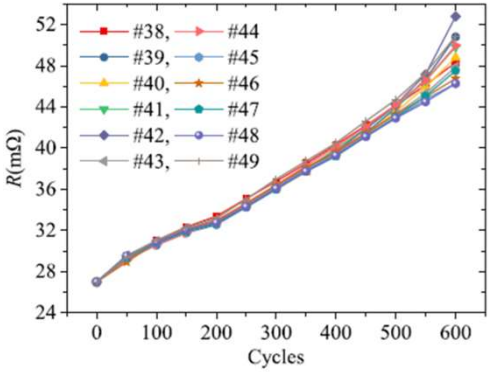
Cell Mismatch: Impedance, Temperature, and How to Fix It



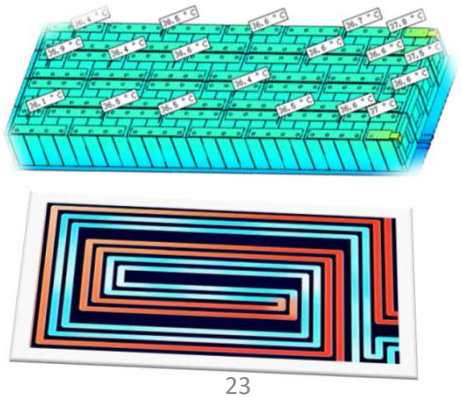
Weakest cell



Weak cell has deeper DOD each cycle and generates more heat.



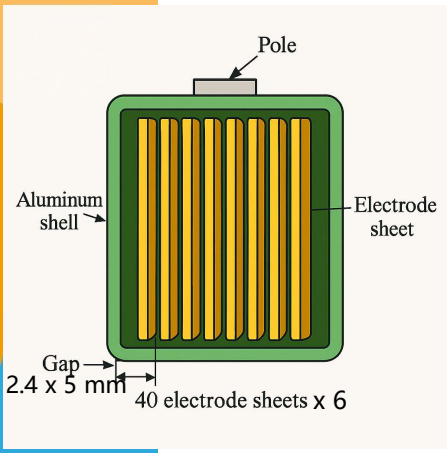
Cell impedance difference becomes larger with cycles.



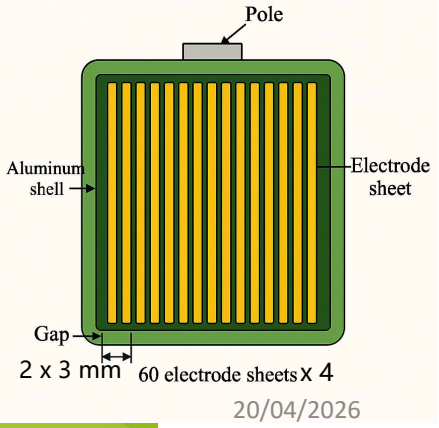
Ambient temperature	25ºC
Average temperature on large surfaces of cells (°C)	34.2ºC
Cell temperature difference	<2.2ºC

Dual-Loop heat dissipation design

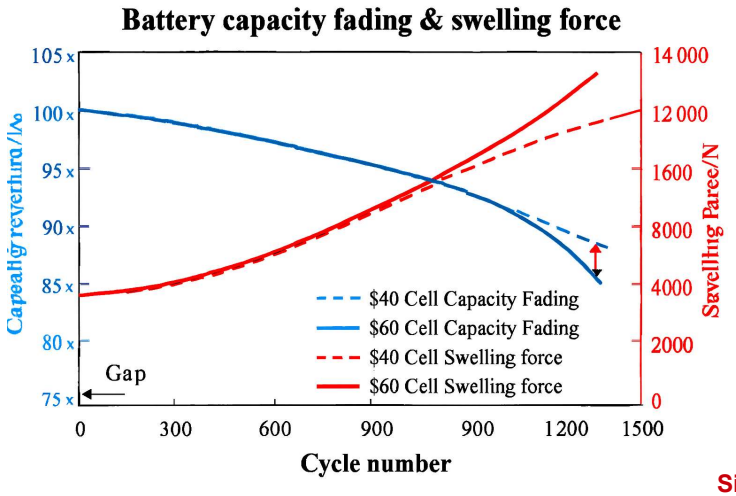
Swelling Force: A Hidden Driver of Degradation



Same electrode sheet: 240

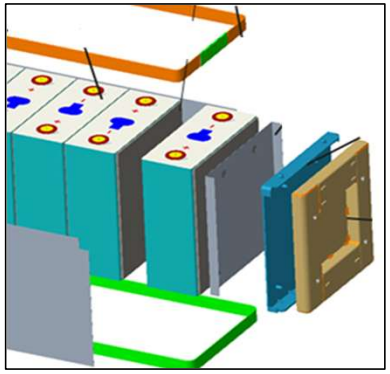


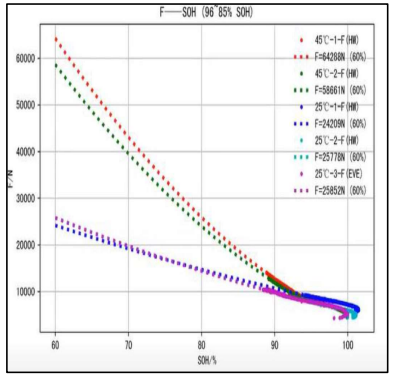
Different layers, insultation covers, lead to different swelling force.



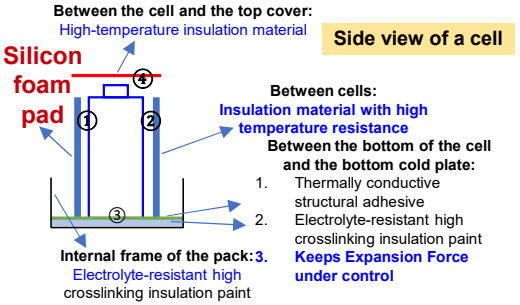
a) Increases with the number of cycles.
b) Gap of cell decay under different swelling force
c) The greater force, the faster the attenuation.

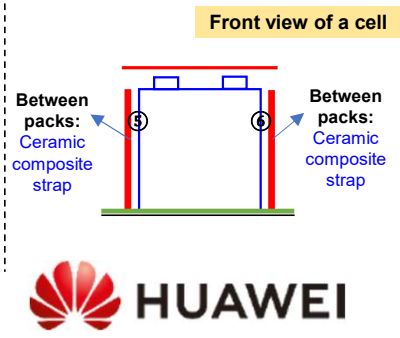
HUAWEI developed swelling force prediction model to optimize the pack structure design and system SOH decay prediction





Swelling force prevention with cell 360º enclosure





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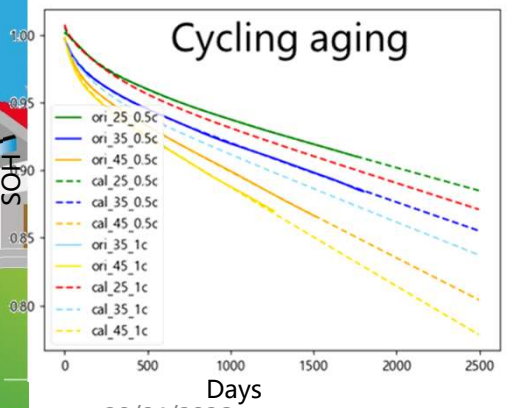
Reliability Testing for Robust SOH Modeling

Cycling Tests

1. 25℃ 1C/1C cycling
2. 35℃ 1C/1C cycling
3. 45℃ 1C/1C cycling
4. 25℃ 0.5C/0.5C cycling
5. 35℃ 0.5C/0.5C cycling
6. 45℃ 0.5C/0.5C cycling
7. 25℃ 0.5CP/0.5CP cycling
8. 35℃ 0.5CP/0.5CP cycling
9. 45℃ 0.5CP/0.5CP cycling

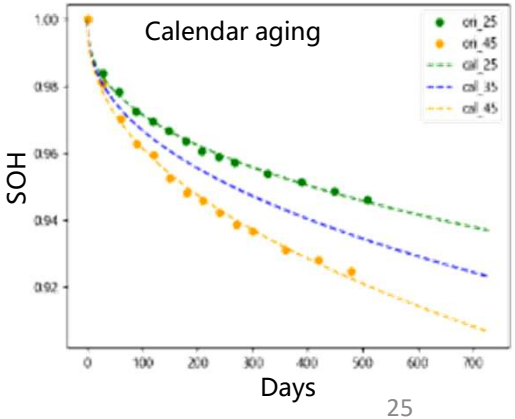
...



Calendar Aging Tests

1. 25℃ 100%SOC Storage
2. 45℃ 100%SOC Storage
3. 60℃ 100%SOC Storage
4. 25℃ 50%SOC Self-discharging
5. 45℃ 50%SOC Self-discharging
6. X℃ X%SOC Self-discharging
7. Float charging
8. +3,000 real cycles

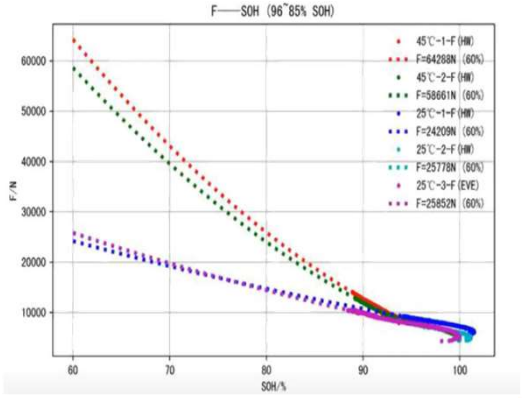
...



Swelling Force Tests

1. 25℃ 1C/1C expansion force
2. 35℃ 1C/1C expansion force
3. 45℃ 1C/1C expansion force
4. 25℃ 0.5C/0.5C expansion force
5. 35℃ 0.5C/0.5C expansion force
6. 45℃ 0.5C/0.5C expansion force
7. 45℃ 0.5CP/1CP expansion force

...



Long Term Tests

1. Continuous cycling over 3000 cycles
2. Battery leakage test
3. Calendar aging over 6 months
4. Cycling under high compress force
5. Lithium plating after cycling

....

Lithium plating after cycling

SOC Accuracy as the Foundation of SOH

Pack-level automatic SOC calibration, higher SOC precision, charging and discharging accurately, higher SOH precision

SOC Calculation

① Ampere-hour integral + accumulated discharge capacity.
② Voltage observer principle for SOC Calibration.
③ Voc for initial SOC estimation

High precision voltage sampling

Extremely high-precision voltage sampling (3mV / 3-10s)

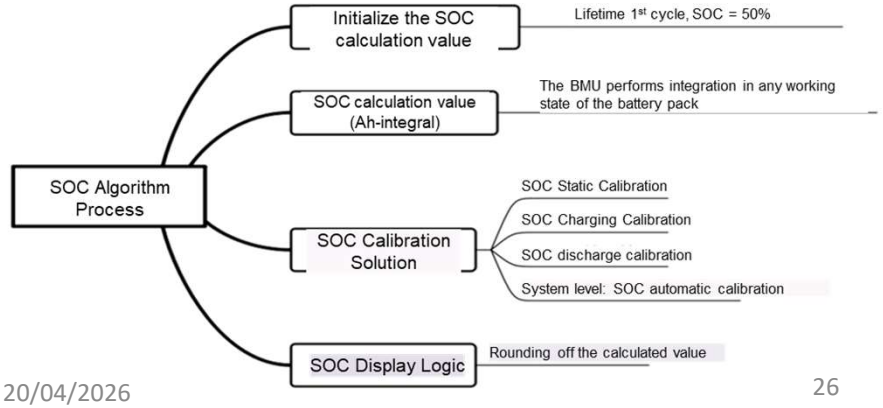
Battery management Integrated chip

Dedicated battery management chip, higher computing power
Cell-level automatic equalization

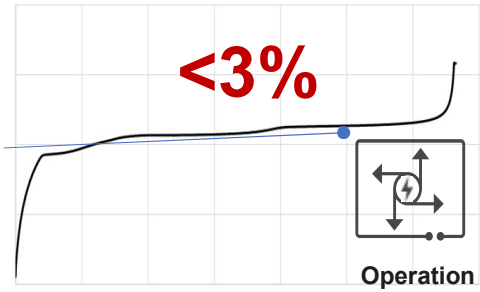
Self-learning algorithm

High-precision battery model modeling, parameter identification, feature extraction

SOC Algorithm Process



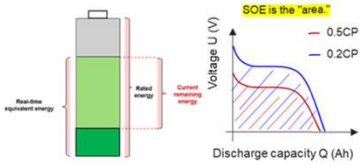
High-precision automatic SOC calibration



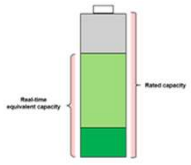
Accurate SOC powers reliable SOH

From Theory to Certification: SOC Accuracy Proven

SOX Calibration **(Accuracy)**

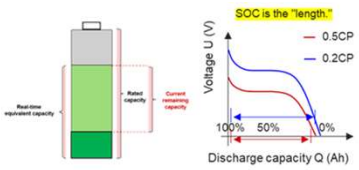


State of Energy (SOE)

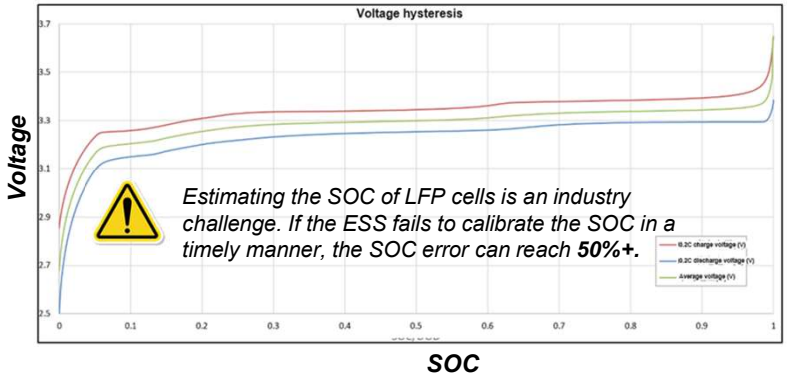


State of Health (SOH)

SOC Balancing **(Consistency)**



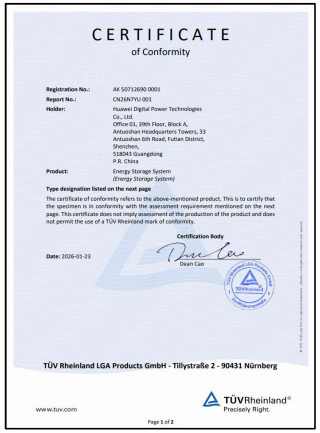
State of Charge (SOC)



Tested according to: 2 PfG 3165/12.25

Identification: Type Designation
LUNA2000-4472-2S, LUNA2000-2236-1S,
LUNA2000-5015-2S, LUNA2000-5015-8S,
LUNA2000-2507-2S

Remark：
Complies with Lv3 in the 2 PfG 3165/12.25
Lv1: Manual calibration assistance
Lv2: Condition-constrained automatic calibration
Lv3: High-precision automatic calibration



3 Unlocking Value: Operational Advantages of Accurate SOC

Eliminating the need for site visits

Lower OPEX

Automatic SOC Calibration

Safety

Avaliability

Reducing Contingency

Accuracy & Consistency

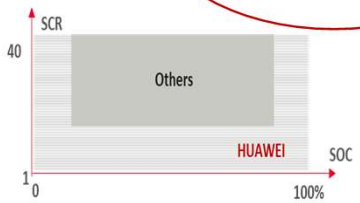
GFM @SCR & DOD full range

RTE

Optimizing CAPEX

Constant Power at 100%

Mismatch Mgmt



Availability > 99.9%

Refined management
Modular design and fewer vulnerable parts

Higher RTE: 91.55%

PCS 800 V AC, higher voltage and less loss

> 2% Higher Capacity

Case study: 500MW - 0.25C

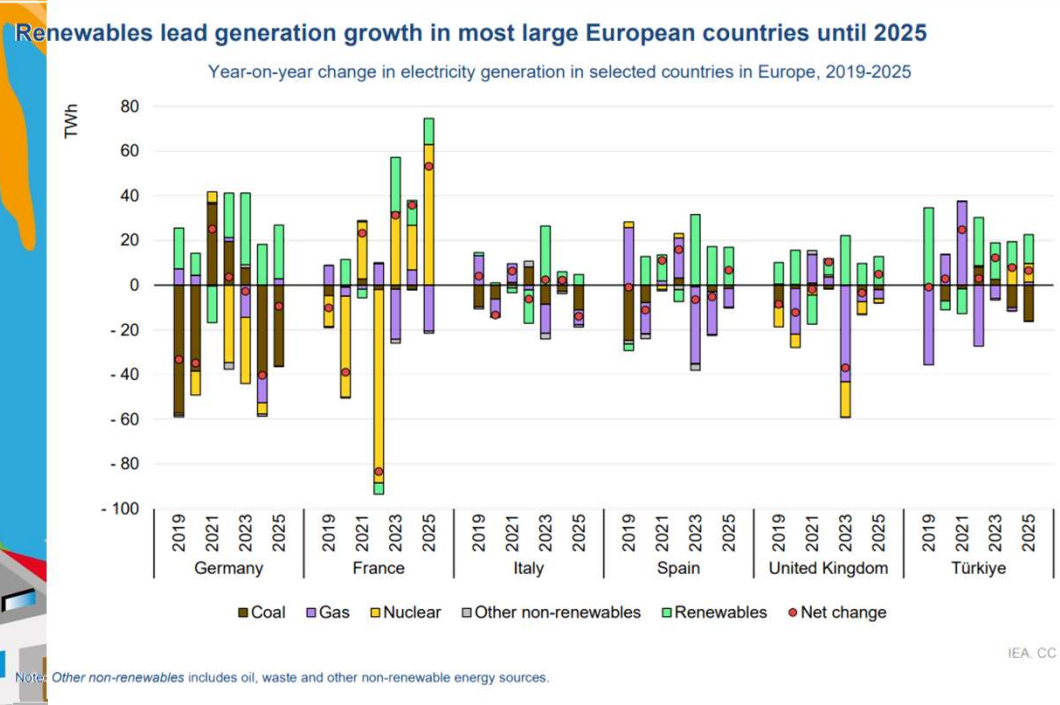
BESS LCOS Calculator	HUAWEI	VENDOR B
ESS Model	LUNA2000-5015-2S	XXXX
Installed rated capacity (MWh)	2,221.65	2,430.00
Installed Power (MW)	555.28	607.50
C-Rate	0.25	0.25
Economic Investment Horizon (years), lifespan	**20**	**20**
CAPEX gap (%), respect HUAWEI		**-18.07%**
IRR Target, pre-tax (% p.a.)	**7%**	
ratio BOS (€/MWh)	**116 €**	**90 €**
O&M + Sust. Capex (% of Upfront Costs p.a.), not representative in this case, see below	**1.81%**	**2.20%**
Depth of discharge (% of total storage that can be cycled)	100%	**95%**
Cycles per day (CPD)	**1.5**	**1.5**
Roundtrip efficiency (%) [@POC RTE (with auxiliary efficiency)] 25ºC	**89.46%**	**88.00%**
Battery Mismatch	**0.0%**	**-2.0%**
BESS Availability (without STS)	**99.0%**	**98.0%**
Average purchase price when charging (€/MWh)	**60**	
Electricity purchase costs (€/year)	**77,963,664 €**	**81,011,646 €**
LCOS, per discharged MWh (€/MWh)	**149.87 €**	**158.97 €**
LCOS gap	**-6%**	

1. Huawei Digital Power: Updates & Trends
2. Digitalization as an improvement of LCOS
3. **Grid forming**
4. Intelligent O&M

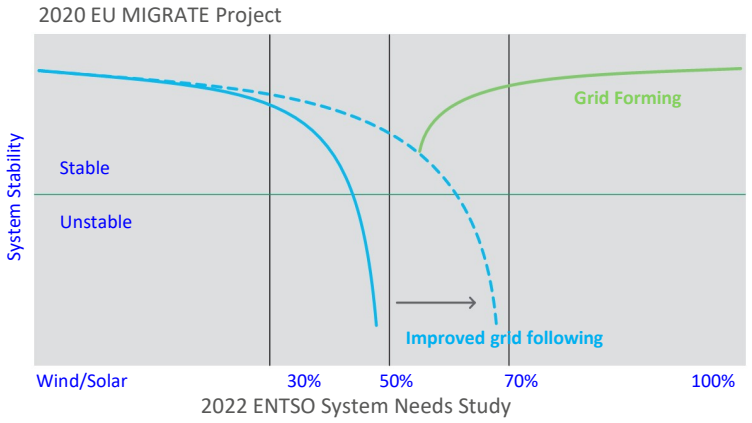
Why “grid forming”?

Renewable industry is booming, while multiple challenges still remain

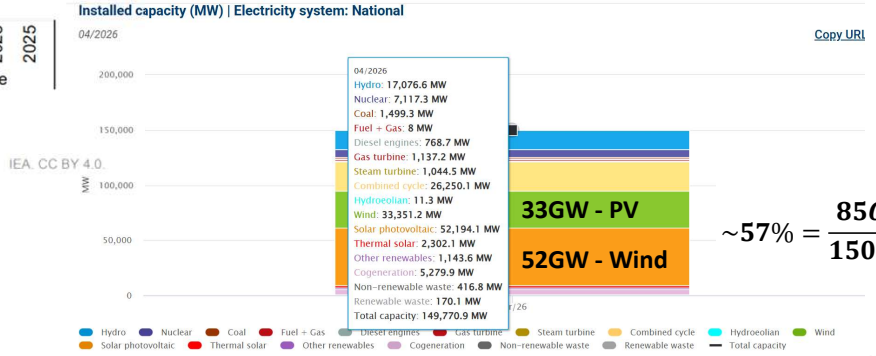
EU considers grid forming as a key to high wind/solar penetration.



Source: 2023 Electricity Market Report IEA (International Energy Agency)



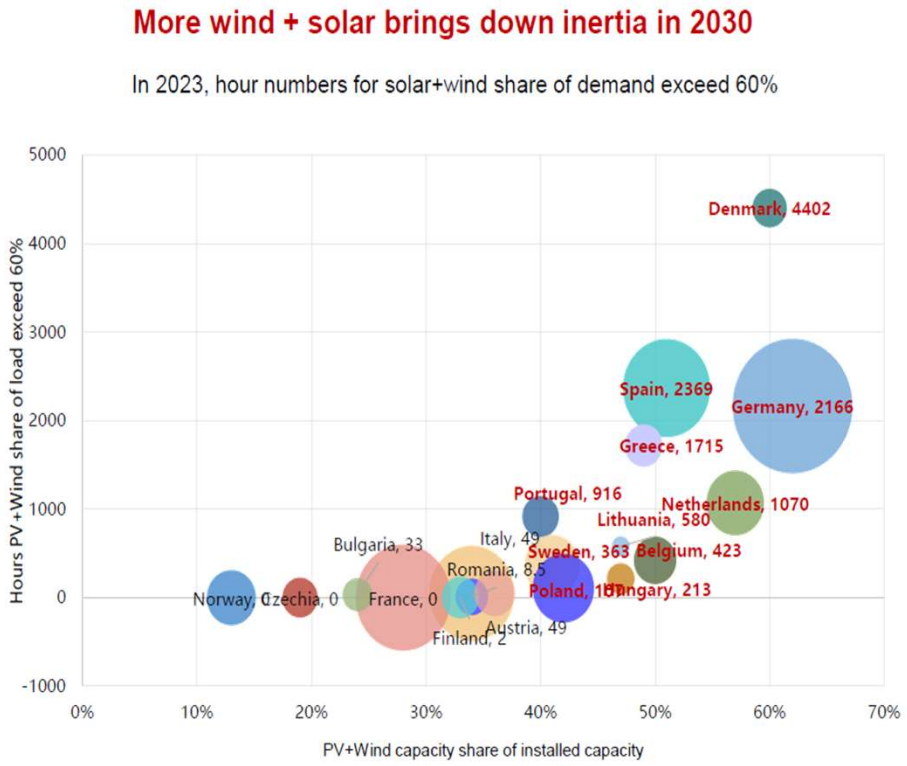
2022 ENTSO System Needs Study

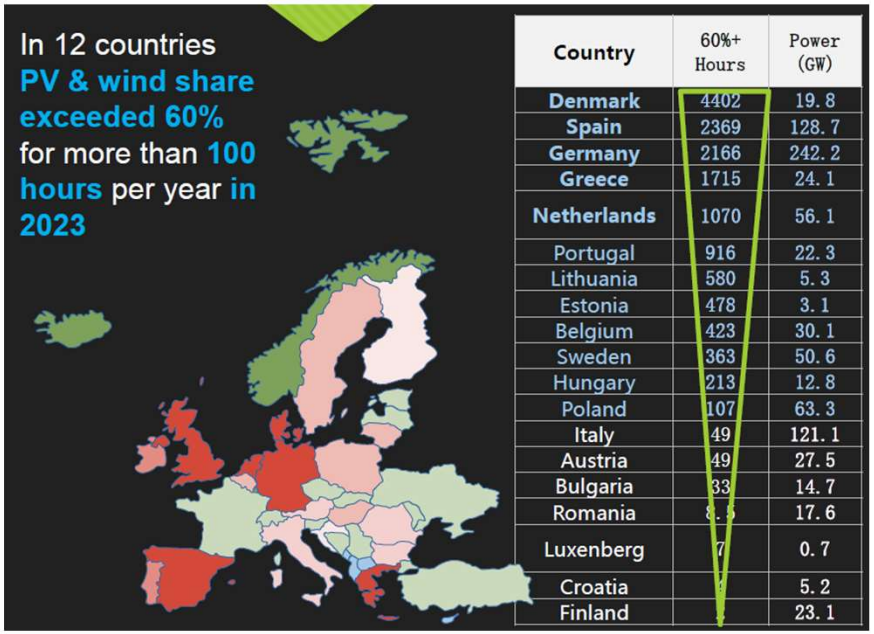


$$\sim 57\% = \frac{85GW}{150GW}\ \%$$

Source: March 2024, Red Eléctrica de España (REE))

Renewable industry is booming, while multiple challenges still remain



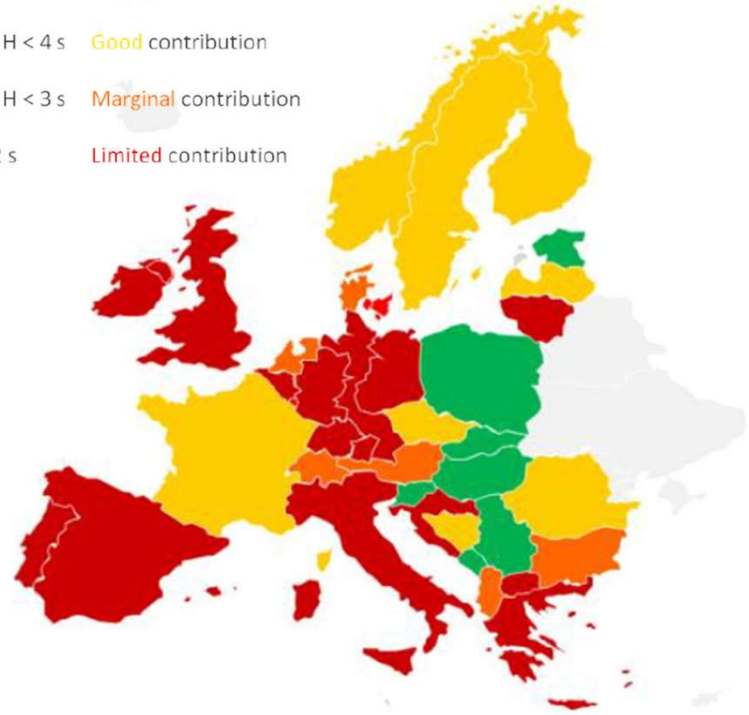


Country	60%+ Hours	Power (GW)
Denmark	4402	19.8
Spain	2369	128.7
Germany	2166	242.2
Greece	1715	24.1
Netherlands	1070	56.1
Portugal	916	22.3
Lithuania	580	5.3
Estonia	478	3.1
Belgium	423	30.1
Sweden	363	50.6
Hungary	213	12.8
Poland	107	63.3
Italy	49	121.1
Austria	49	27.5
Bulgaria	33	14.7
Romania	8.5	17.6
Luxenberg	7	0.7
Croatia	5	5.2
Finland	2	23.1

Weak power system strength has become an issue

Inertia contribution colouring code:

- Green H ≥ 4 s Very good contribution
- Yellow 3 s ≤ H < 4 s Good contribution
- Orange 2 s ≤ H < 3 s Marginal contribution
- Red H < 2 s Limited contribution



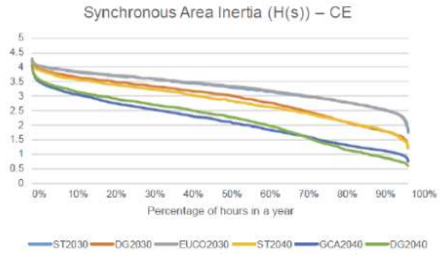
Indicating contribution of each TSO to the TSI constant (source: TYNDP 2016 reflecting 2030 scenario, [7]).

From PNIEC: 81% of the energy in 2030 shall provide from renewable energies

Grid become 'weaker' with renewable penetrations raising

Europe

Entso-E predicts: The system strength of major EU countries will decline rapidly in the next five years.



Inertia constant in the CE system for given percentage of hours per year under various grid development scenarios. *Source: ENTSOE*

Spain

SRAP (Sistema de reducción automática de potencia)

La CNMC aprueba un nuevo procedimiento de operación del sistema eléctrico para la integración masiva de generación renovable

25 Jan 2022 | Energía Energía, Nota de prensa

- La integración masiva de generación renovable en la red eléctrica prevista en los próximos años exige adaptar los procedimientos de resolución de congestiones en la red.
- Esta adaptación requiere un nuevo procedimiento para la operación del sistema eléctrico, un herramienta para la reducción automática de la potencia de la generación.
- El nuevo sistema permitirá la entrada máxima de generación renovable garantizando la seguridad del suministro y el uso óptimo de las redes.

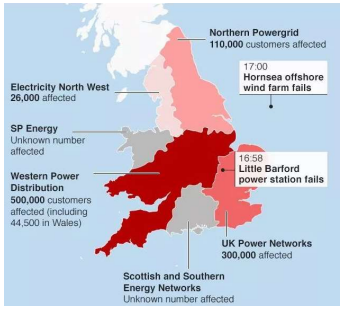
La CNMC ha aprobado un nuevo procedimiento de operación del sistema eléctrico peninsular (P.O.3.11 y modificación del P.O.3.2) para desarrollar un sistema de reducción automática de potencia (**DCOOR/DE/007/21**).

La incorporación de este sistema en el proceso de resolución de restricciones técnicas permitirá incrementar la integración de generación renovable, optimizando el uso de las redes y maximizando la integración de generación renovable.

New Energy Makes Grid Weaker, Transient Issues Come to The Fore

1 Weak inertia support and frequency instability

The Blackout in UK 2019



The failure of Little Barford power plant caused power loss of 730 MW

System frequency starting to decrease

Horsea offshore wind farm was largely off the grid, with power fed into the grid reduced by 900 MW.

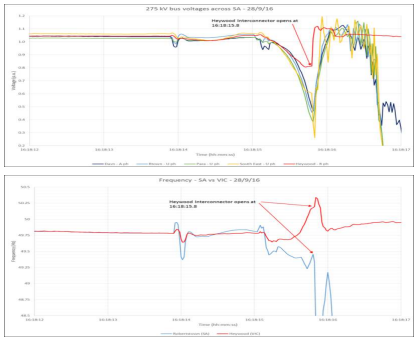
System frequency dropped to 48.9Hz

beyond the allowed range of 49.8–50.2 Hz and triggered mass load shedding

2 Fault ride-though failure on weak networks

Australia's power blackout in 2016

- On Sept. 28th, 2016, Wind farm tripped caused by typhoons and rainstorms.
- Voltage drops occurred for 6 times and caused voltage and frequency collapse. South Australia suffered a 50-hour blackout.

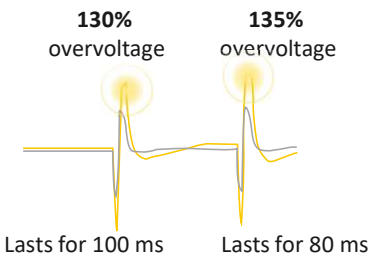


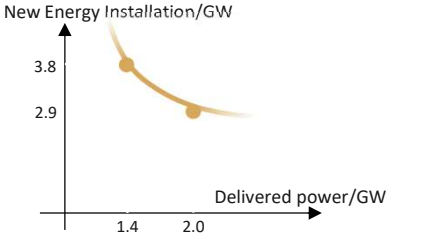
Voltage collapse

Frequency collapse

3 High proportion of renewable difficult to deliver

HVDC transient overvoltage problem (Qinghai-Henan DC transmission)

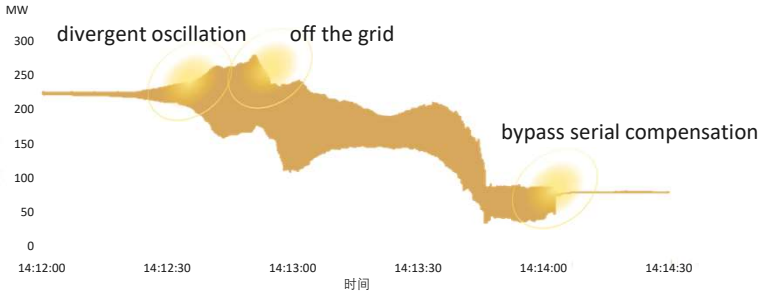




New energy installation and delivery capacity are negatively related!

4 Wide-band oscillation

A wind power station in Guyuan, Hebei, China



The Whole Industry Address Those Challenges Actively, Still with a Number of Restrictions

Diesel generator

Environmental pollution, high LCOE, and complex maintenance

Synchronous condenser

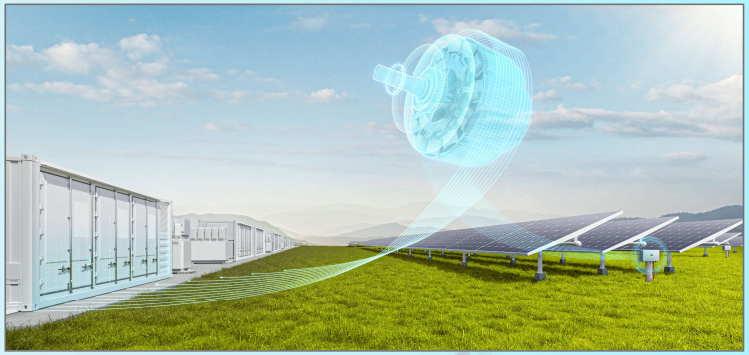
24-hour manual attendance of rotating equipment and regular on-site O&M

New construction or renovation of thermal generators

Restriction on new construction and environmental pollution

PV + ESS Generator

Combine PV and Energy Storage with Intelligence, Accelerating PV to be Primary Energy Source



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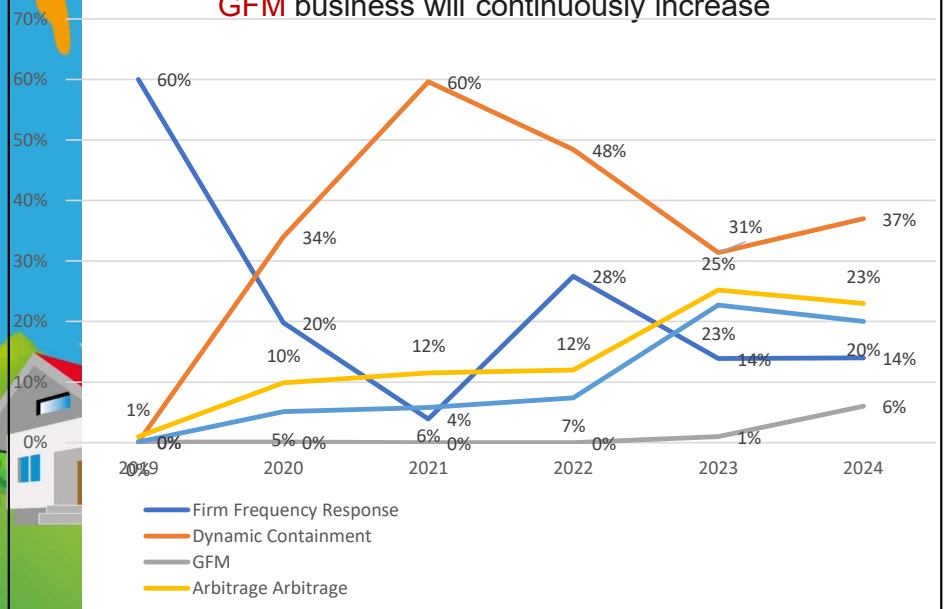
One platform matching multiple business mechanism, continuous creating value for customer

UK ESS Profit Model Evolution from 2019 to 2024

Frequency Reserve Business fluctuate shapely

Arbitrage business will exist for long term

GFM business will continuously increase



*Source of Revenue of UK Gresham House

All Business, Full life time, Higher Revenue

1 platform→All business model

Energy market	Capacity market	FM market	GFM Market	PPA	Long-term energy	Microgrid

SCL	Inertia	Black start	1st /2nd /3rd FM	Wideband oscillation suppression	Reactive overload capacity	Constant power capability	SOC accuracy	Availability	On and off grid
1.5/3/6 times VS 3 times	3-20s VS 20s	Mins level VS day level	< 100 ms VS >200ms	0.1~100HZ VS 0.2~2.5HZ	1.5 times VS 1.2 times	0~100% VS 0~94%	3% VS 5%	99.9% VS 98%	One platform support

Percentage of renewable energy < 40%	Percentage of renewable energy < 60%	Percentage of renewable energy ~ 100%
1 pcs Smart PCS	2 pcs Smart PCS	3~N pcs Smart PCS

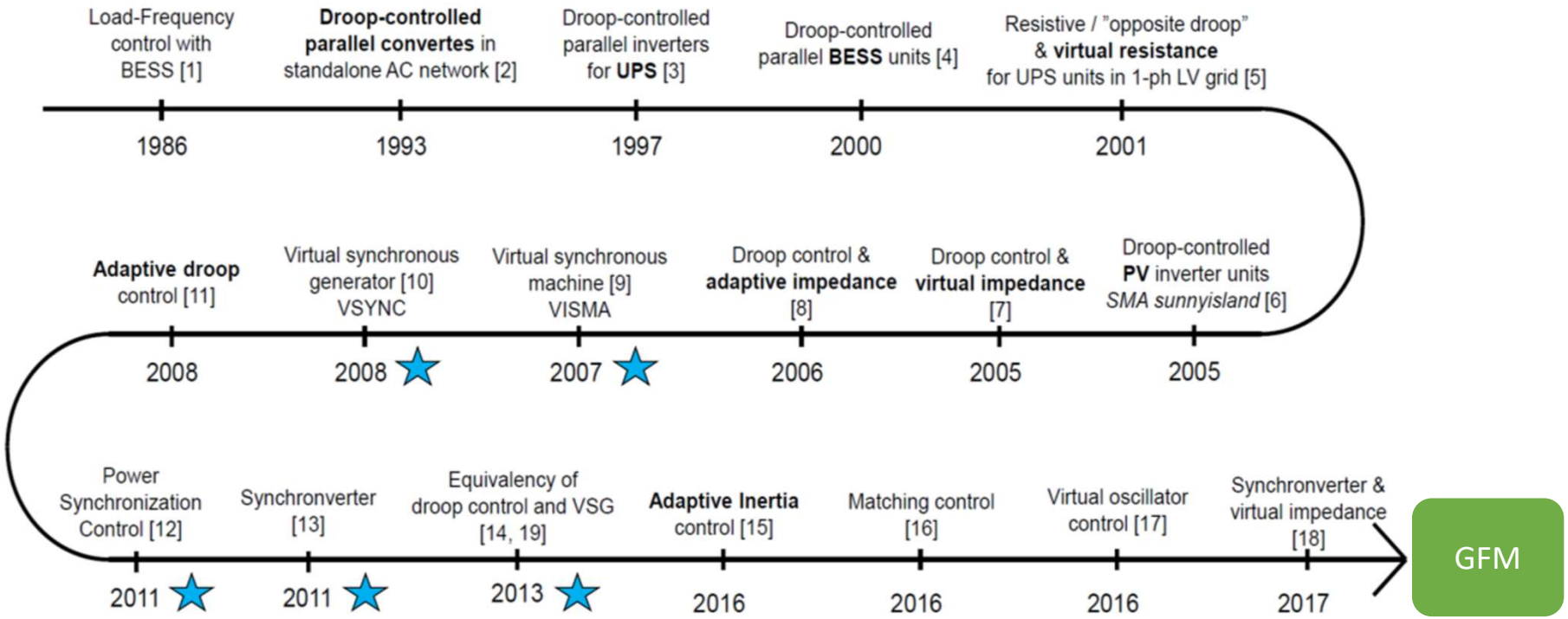
Grid-forming capability ready- Continuous upgrade

HUAWEI

Grid Forming converters

Change in operating paradigm

Grid Forming for Renewable Power Generation - Academic Research



Key metrics to understand localized power system strength & resiliency

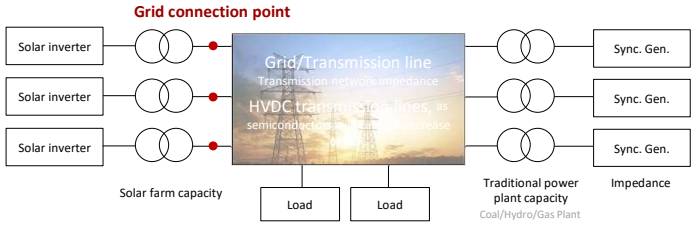
1 Short Circuit ratio (SCR)

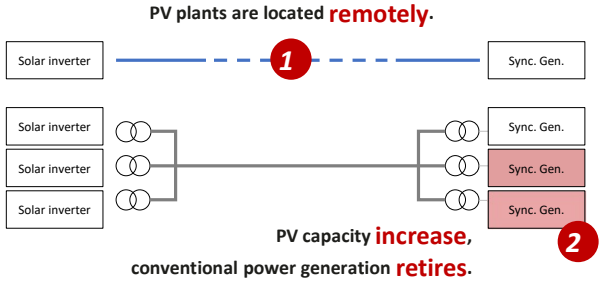
Increased Renewable Energy Penetration Rate equals to Weakened Grid Strength (SCR)

Short circuit ratio (SCR)

SCR is the ratio of the short-circuit capacity at the grid connection point to the rated capacity of a power plant. It measures the impact of power injection on the voltage quality and stability of local grids, and reflects the grid strength.

A higher SCR indicates a stronger grid and grid connection ability.





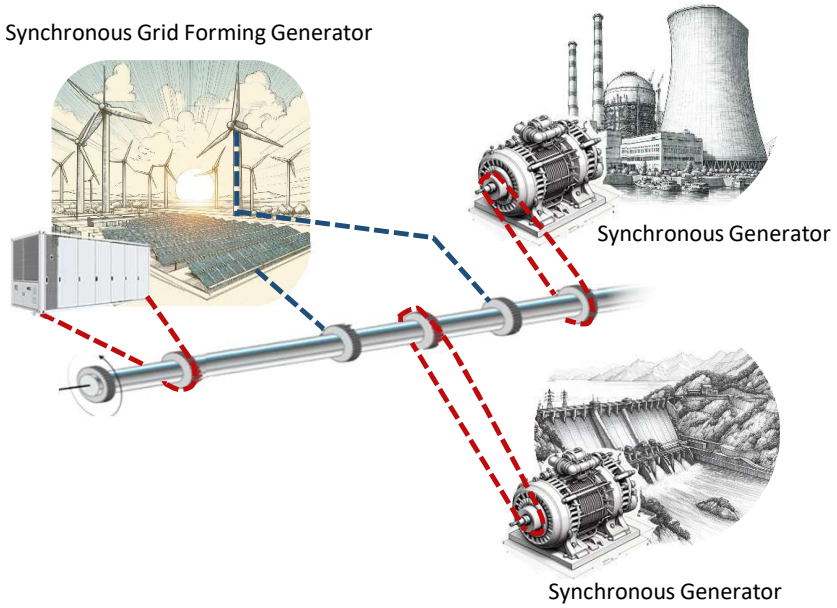
$$SCR = \frac{\text{Square of Voltage at the grid connection point}}{\text{Equivalent impedance at the grid connection point}} \Big/ \text{Solar Farm Capacity}$$

1. **Longer** power transmission lines result in **larger equivalent impedance** at the grid connection point.
2. **Less capacity of** traditional power plants result in **larger equivalent impedance** at the grid connection point.

...

Key metrics to understand localized power system strength & resiliency

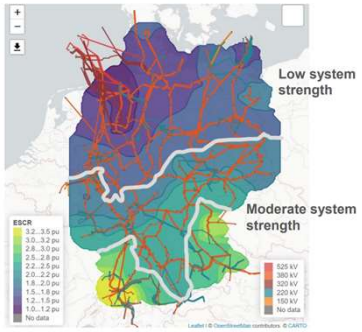
1 Inertia H (s)



Key metrics to understand localized power system strength & resiliency

1 Short Circuit ratio (SCR)

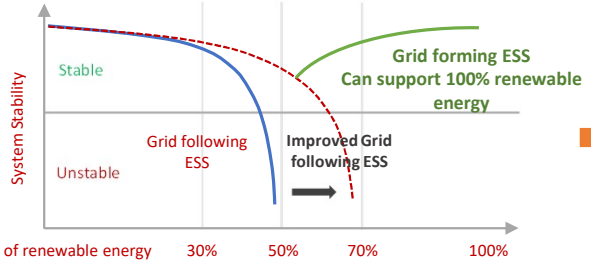
SCR> 3	Strong grid (~Voltage stability and power reserves)
3 > SCR > 2	Weak grid (~Voltage instability and control problems)
SCR < 2	**Very weak**



Source: VDE FNN

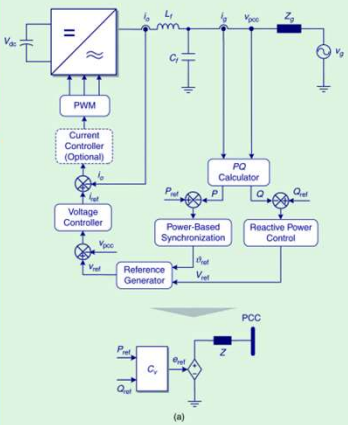
2 Inertia H (s)

Limited	Marginal	Good	Very Good
0-2 s	2-3 s	3-4 s	≥4 s

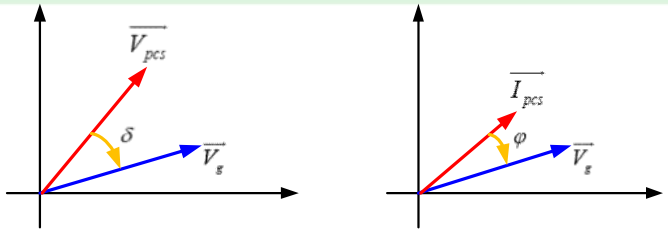


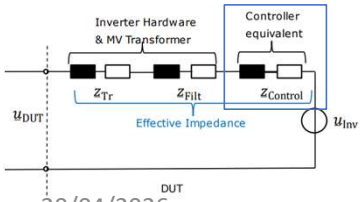
Power system stability and control by Kundur:

"The ability of an electric power system, for a given initial operating condition, to regain a state of operating equilibrium after being subjected to a physical disturbance, with system variables bounded so that practically the entire system remains intact."

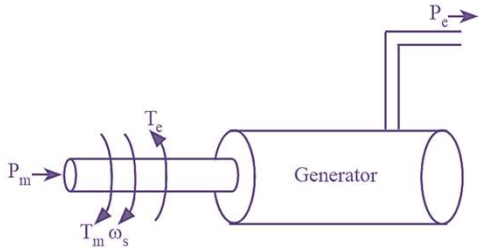


GRID FORMING **GRID FOLLOWING**





In GFM converters, **there is no rotor and no physical mass** rotating, but controllers can emulate the behaviour of a synchronous generator using the **Swing Equation**.



Swing equation

$$\frac{T_j}{w_s}\frac{d\delta}{dt^2} = P_m - P_e$$

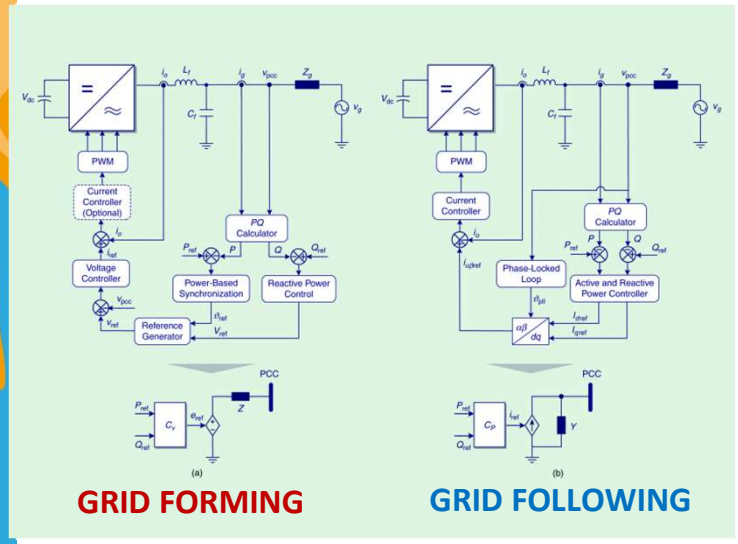
$\theta_m = rotor\ angle$

$T_j = 2H\ Inertia$

$T_m, T_e = mechanical\ and\ electrical\ Torque$

$$J\frac{d^2\theta_m}{dt^2} = T_m - T_e$$

$P_m, P_e = mechanical\ and\ electrical\ power$

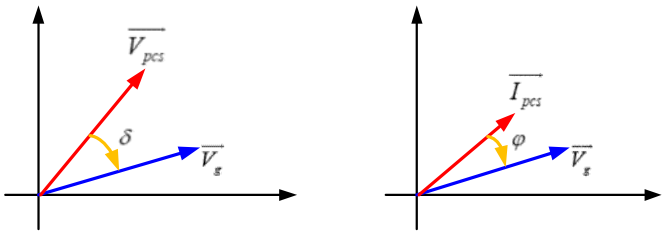


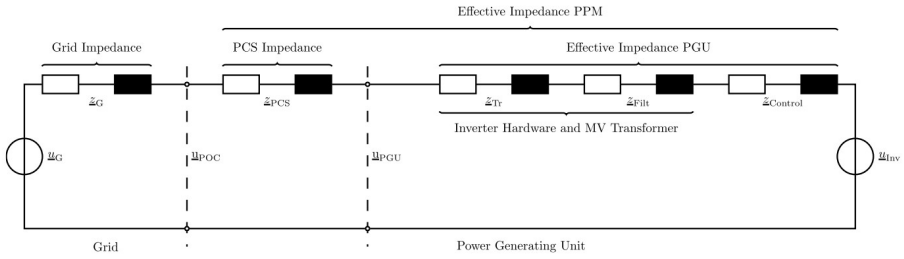
In GFM converters, **there is no rotor and no physical mass** rotating, but controllers can emulate the behaviour of a synchronous generator using the **Swing Equation**.

REE GFM proposal:

7. Where grid forming capability is specified by the relevant TSO in coordination with the relevant system operator in accordance with paragraph 5 or defined in Articles 20, 21 and 22, a power park module shall be capable of providing grid forming capability at the connection point as listed below, considering the sub-cycle character of the physical quantities where appropriate.

(a) Within the power park module's current and energy limits, the power park module shall be capable of behaving at the terminals of the individual unit(s) as a voltage source behind an internal impedance (Thevenin source), during normal operating conditions (non-disturbed network conditions) and upon inception of a network disturbance (including voltage, frequency and voltage phase angle disturbance). The Thevenin source is characterized by its internal voltage amplitude, voltage phase angle, frequency and internal impedance.





Smart PV & ESS Generator: Grid Stability Support Capacity Equal to Synchronous Generator

		Synchronous generator	Huawei Grid Forming	Comp
Voltage Support	1、*Transient Short Circuit Current Amplitude*	3 p.u.	**3 p.u / 1.2 p.u	=
	2、*Transient Short Circuit Current Startup time*	< 5ms	< 5ms	=
	3、*Transient Short Circuit Current response time*	< 20ms	< 10ms	+
	4、*Transient Short Circuit Current duration*	20s	10s (Limited by transformer capacity)	-
Frequecy /Angle Support	1、*Inertia time constant*	Hydraulic1~3s，Thermal 6~9s	2 ~ 20s Settable	+
	2、*Primary/Secondary Frequency Regulation*	Yes	Yes	+
	3、*Phase angle jump suppression*	Yes	Yes	=
	4、*Inertia support response time*	< 5ms	< 5ms	=
Others	1、*Oscillation suppression*	Yes	Yes（broadband）	+
	2、*Power Quality*	THDu < 1%	THDu < 1.5%	-
	3、*Blackstart*	Yes（>10 min）	Yes（< 1 min）	+
Economy	1、*Equipment investment cost*	—		~
	2、*O&M Cost*	—		+
	3、*O&M Difficulty*	24H	Unattended	+
	4、*Life Time*	30Y	20Y	-

Smart PV Controller
SUN2000-506KTL

Accurate Grid Sensing
+
Advanced Control

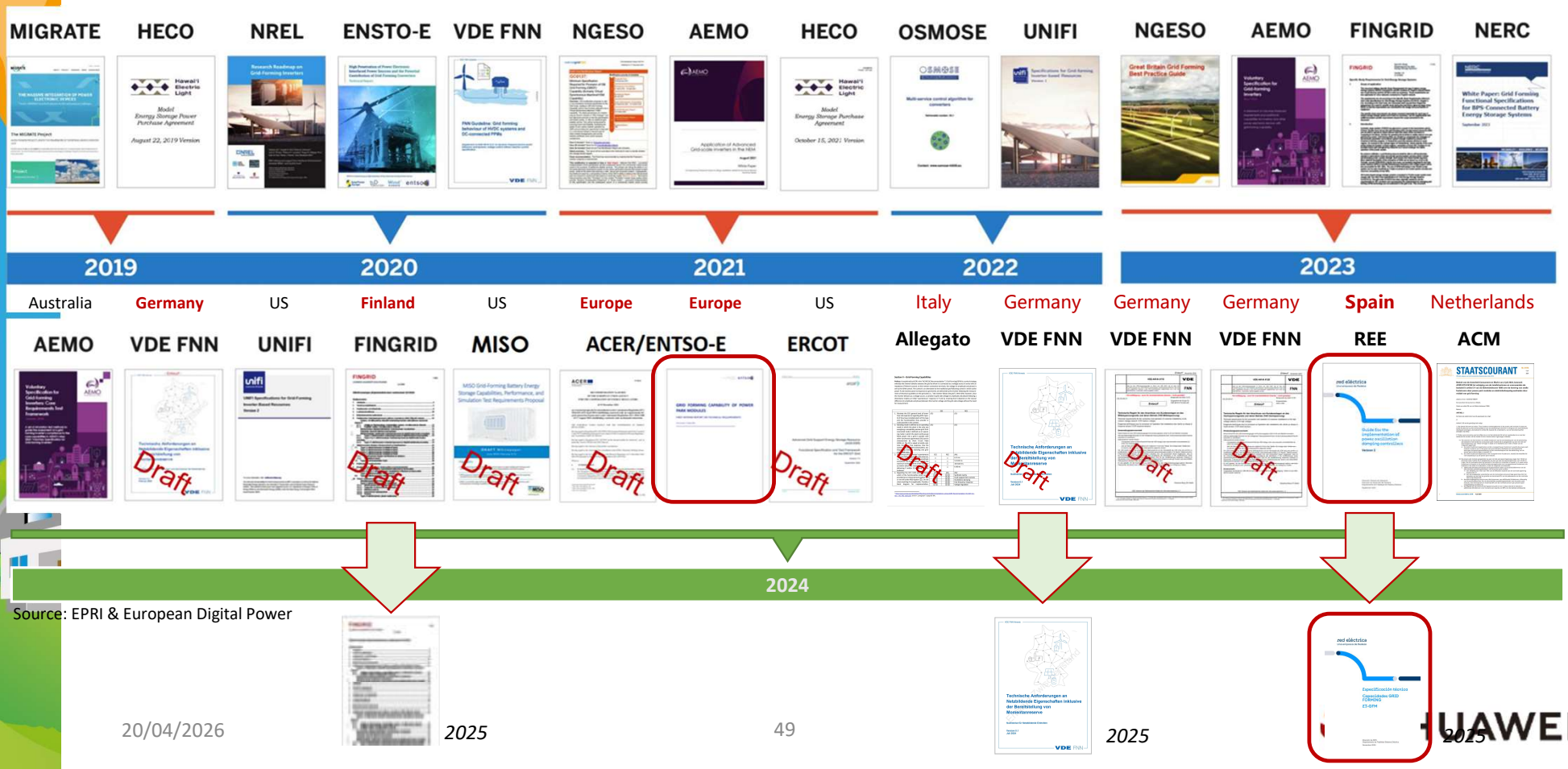
Smart PCS
LUNA2000-213KTL

Safe & Reliable — **Grid Forming** — **Full Intelligence** — **Optimal LCOE**

GFM Testing Practice

Adapting to Spanish market

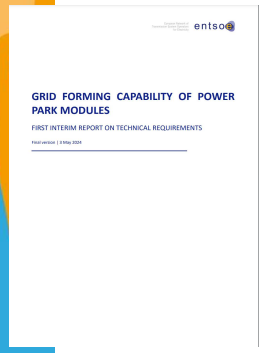
From 2024, GFM Standardization Began to Accelerate



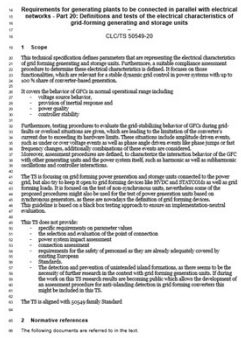
Source: EPRI & European Digital Power

Next Step in Europe: Implementation Guidance Document for RfG 2.0

ENTSO-E Requirement



EN SO 74-1: Testing procedure


Requirements for generating plants to be connected in parallel with electrical networks - Part 20: Definitions and tests of the electrical characteristics of grid-forming generating and storage units

CLC/TS 50549-20

1 Scope

This technical specification defines parameters that are representing the electrical characteristics of grid forming generating and storage units. Furthermore, a suitable compliance assessment procedure to determine these electrical characteristics is defined. It focuses on those functionalities, which are relevant for a stable dynamic grid control in power systems with up to 100 % share of converter-based generation.

It covers the behavior of GPUs in normal operational range including
- voltage source behavior,
- provision of inertial response and
- power quality,
- controller stability.

Furthermore, testing procedures to evaluate the grid-stabilizing behavior of GPUs during grid-faults or overload situations are given, which are leading to the limitation of the converter's current due to exceeding the converter limit. Typical grid-faults are e.g. voltage drops events, such as under or over voltage events as well as phase angle drops events like phase jumps or fast changes of the phase angle. Also frequency events are considered in this TS.
Moreover, assessment procedures are defined, to characterize the interaction behavior of the GPU with other generating units and the power system itself, such as harmonic as well as subharmonic oscillations and controller interactions.

The TS is focusing on grid forming power generation and storage units connected to the power grid, but also try to keep it open to grid forming devices like HVDC and FLEXTOXIA as well as grid forming loads. It is focused on the test of non-synchronous units, nevertheless some of the proposed procedures might also be used for the test of power generation units based on synchronous generators, as these are nowadays the definition of grid forming devices.
The definition is based on a black box testing approach to ensure an implementation-neutral evaluation.

This TS does not provide:
- specific requirements or parameter values
- rules for the application or the point of connection
- power system impact assessment
- connection assessment
- requirements for the safety of personnel as they are already adequately covered by existing European
- Directives
- The detection and prevention of unintended island formations, as there seems to be the necessity of further research in the context with grid forming generation units. If during the work on this TS research results are becoming public which allows the development of an assessment procedure for anti-islanding detection in grid forming converters this might be included in this TS.

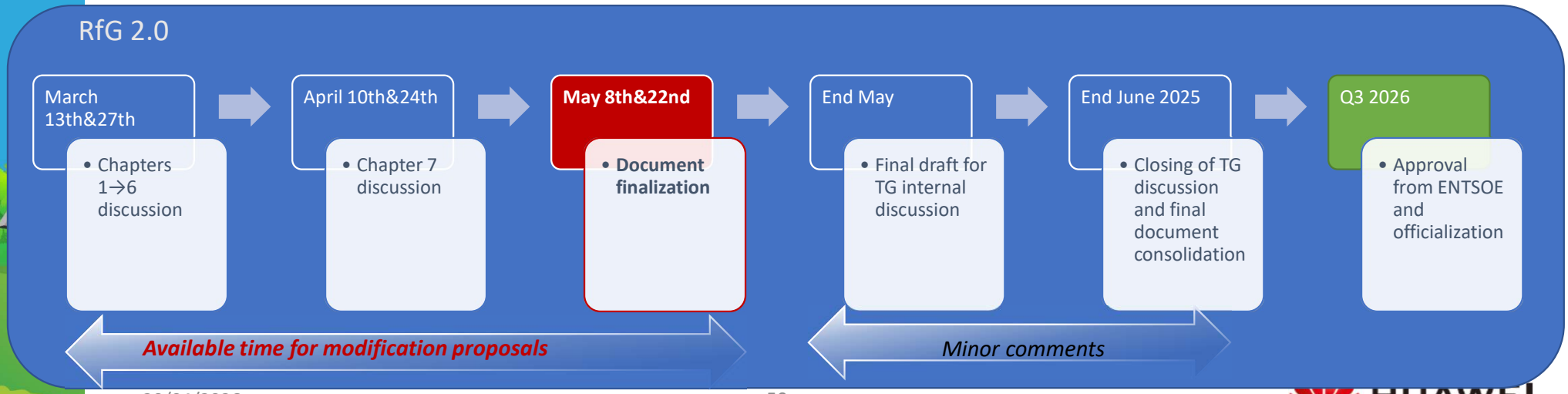
This TS is aligned with 50549-family Standard

2 Normative references

The following documents are referred to in the text.

Verifying compliance with the proposed requirements with the following tests:

- Voltage phase angle change
- RoCoF
- Symmetrical and asymmetrical voltage change
- Loss of last synchronous generator



Final implementation into Grid Codes:

Spanish NTS

RED ELÉCTRICA DE ESPAÑA
Grupo Red Eléctrica
aeléc
ufd Grupo Naturgy
cide
aniversario E50 ASEME

Technical standard for monitoring the compliance of power generating modules according to EU Regulation 2016/631

Review	Reason	Date	Comments
1.0	Publication	18/7/2019	
2.0	Publication version 2	3/11/2020	Adoption of Order TED/749/2020 and Royal Decree 647/2020. English version contains Corrigendum to Spanish version 2 published on 13/4/2021.
2.1	Publication version 2.1	9/7/2021	Relevant modifications to version 2

Spain Grid Code: Orden TED 749

BOLETÍN OFICIAL DEL ESTADO
Núm. 208 Sábado 1 de agosto de 2020 Sec. I. Pág. 62406

I. DISPOSICIONES GENERALES

MINISTERIO PARA LA TRANSICIÓN ECOLÓGICA Y EL RETO DEMOGRÁFICO

8965 *Orden TED/749/2020, de 16 de julio, por la que se establecen los requisitos técnicos para la conexión a la red necesarios para la implementación de los códigos de red de conexión.*

I

El Reglamento (UE) 2016/631 de la Comisión, de 14 de abril de 2016, que establece un código de red sobre requisitos de conexión de generadores a la red, define los requisitos técnicos para la conexión a la red de las instalaciones de generación de electricidad.

El Reglamento (UE) 2016/1388 de la Comisión, de 17 de agosto de 2016, por el que se establece un código de red en materia de conexión de la demanda, define los requisitos de conexión a la red de instalaciones de demanda y de distribución conectadas a la red de transporte, de las redes de distribución, incluidas las redes de distribución cerradas, y de las unidades de demanda utilizadas por una instalación de demanda o una red de distribución cerrada para prestar servicios de respuesta de demanda a los gestores de red y a los gestores de red de transporte pertinentes.

El Reglamento (UE) 2016/1447 de la Comisión, de 26 de agosto de 2016, por el que se establece un código de red sobre requisitos de conexión a la red de sistemas de alta tensión en corriente continua y módulos de parque eléctrico conectados en corriente continua, define los requisitos para la conexión a la red de sistemas de alta tensión en corriente continua (sistemas HVDC) y de módulos de parque eléctrico conectados en corriente continua.

Los citados reglamentos comunitarios constituyen los denominados códigos de red de conexión, los cuales establecen los requisitos técnicos mínimos que deberán cumplir las instalaciones de generación, las de demanda y los sistemas HVDC que se conecten a la red eléctrica, al objeto de garantizar la seguridad de los sistemas eléctricos y favorecer la integración de energías renovables.

Si bien una parte de los requisitos técnicos establecidos en los tres reglamentos comunitarios son de directa aplicación, otros no están completamente detallados y su aplicación requiere que, conforme a lo establecido en los mismos, sean propuestos por los gestores de la red y posteriormente aprobados y publicados por la entidad designada por el Estado miembro, la cual será la autoridad reguladora, salvo disposición en contra de dicho Estado miembro.

Con el fin coordinar las propuestas que, de acuerdo con lo señalado anteriormente, debían presentar los gestores de red en relación con los requisitos no completamente desarrollados en el Reglamento (UE) 2016/631, de 14 de abril de 2016, en el Reglamento (UE) 2016/1388, de 17 de agosto de 2016, y en el Reglamento (UE) 2016/1447, de 26 de agosto de 2016, y proporcionar al mismo tiempo un foro de debate previo donde plantear y resolver aspectos relacionados con el proceso de implementación de dichos reglamentos, en 2016 se crearon, bajo la coordinación del operador del sistema eléctrico, varios grupos de trabajo a los que asistieron, además de los gestores de red, representantes de los agentes afectados, así como del Ministerio para la Transición Ecológica y el Reto Demográfico y de la Comisión Nacional de los Mercados y la Competencia.

Con fecha 29 de mayo de 2018, Red Eléctrica de España, S.A., presentó al actual Ministerio para la Transición Ecológica y el Reto Demográfico una propuesta de modificación del procedimiento de operación 12.2, sobre requisitos mínimos de diseño, equipamiento, funcionamiento, puesta en servicio y seguridad de instalaciones de generación y demanda con conexión a la red de transporte, la cual incluía, entre otras cuestiones, su propuesta relativa a los requisitos técnicos que, de acuerdo con lo señalado

Spain Grid Forming Requirements

red eléctrica
Una empresa de Redeia

Especificación técnica

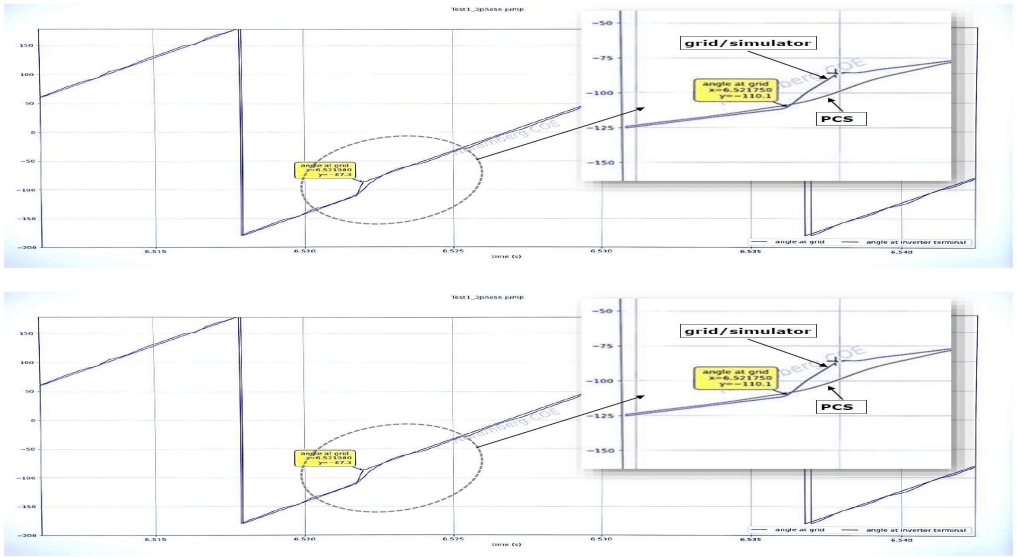
Capacidades GRID FORMING

ET-GFM

Dirección de DDS
Departamento de Fiabilidad Sistema Eléctrico
Noviembre 2025

Voltage angle jump: Disturbance of 30º at grid level

PCS response remains stable without oscillation



REE GFM proposal: "In the event of angular jumps due to network operations, it is proposed that the equivalent requirement of the MGES."

MGES connected to the transmission grid shall be able to withstand the closing of grid lines with an angular difference between poles of the line breaker of 30 º, and occasionally up to 35 º.

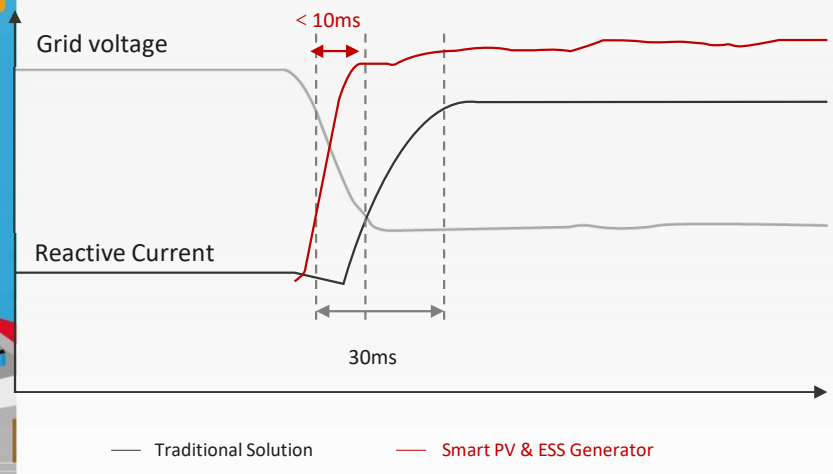
PCS response according to synchronous generator equation:

$$\Delta p_0 = -\frac{1}{x_{\text{Eff}}} \cdot (\sin \delta_1 - \sin \delta_2)$$

Redefine Voltage Stability: Upgraded hardware + Grid Forming algorithm , achieving high reactive current and quickly respond to grid voltage changes

Reactive current to support voltage

Smart PV & ESS Generator : 1.2 times VS Traditional solution: 1.05 times



REE GFM requirements

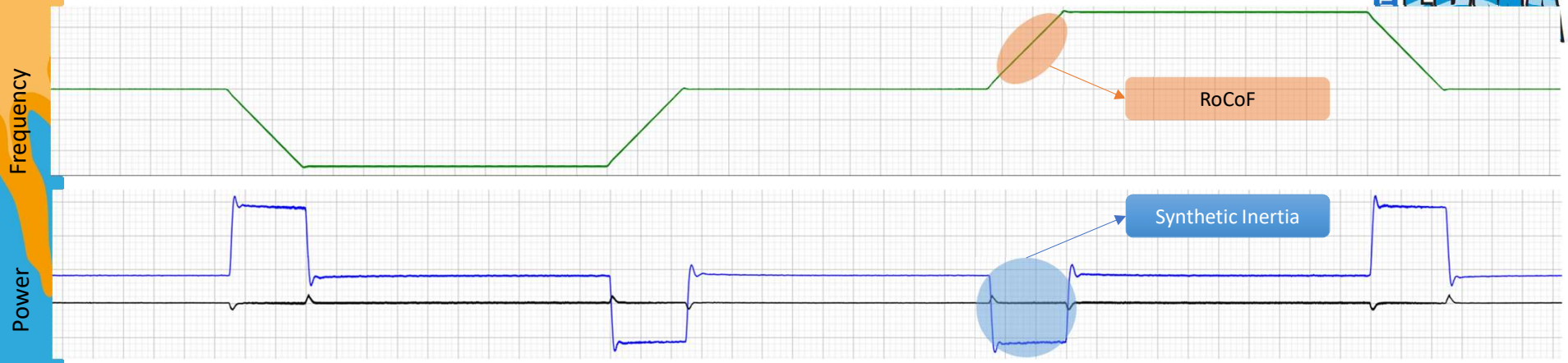
They will not have to comply with the requirement for fast current injection.

if GFM capability is available, fast current injections will be provided naturally and quasi-instantaneously, so such a requirement only makes sense for GFL MPEs. In any case, the **subsequent fast voltage control emulating the action of an AVR is proposed to be required as an additional capability for MPEs GFL.**

They shall be able to withstand voltage sags and transient over-voltages as required for MPE GFL.

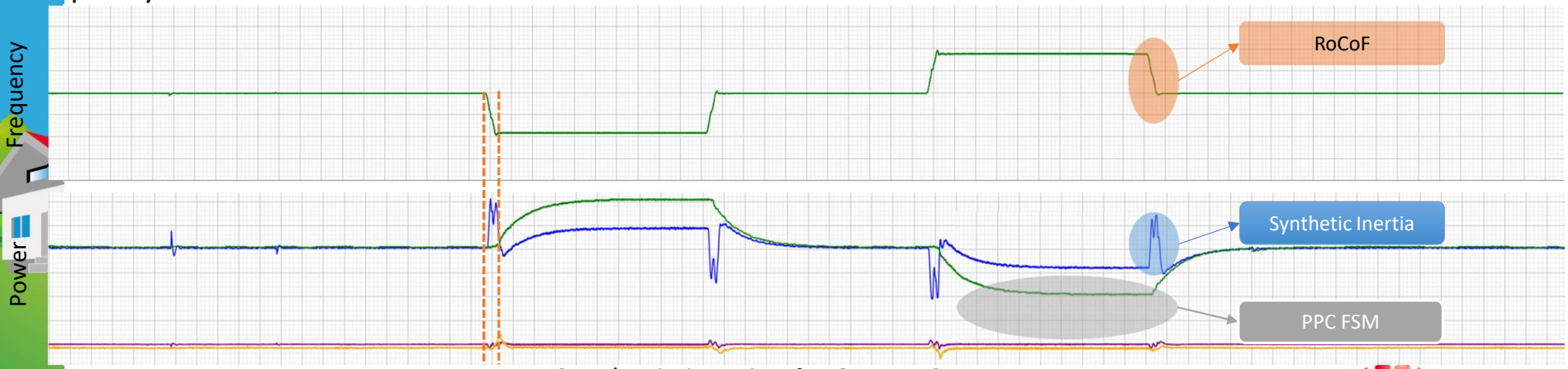
They shall have the technical capability of transiently achieving 1,2 times the rated apparent current for at least 500 ms.

Frequency event: **PCS in GFM mode**



Frequency event: **PCS in GFM mode** + **PV** + **PPC**

PV PCS

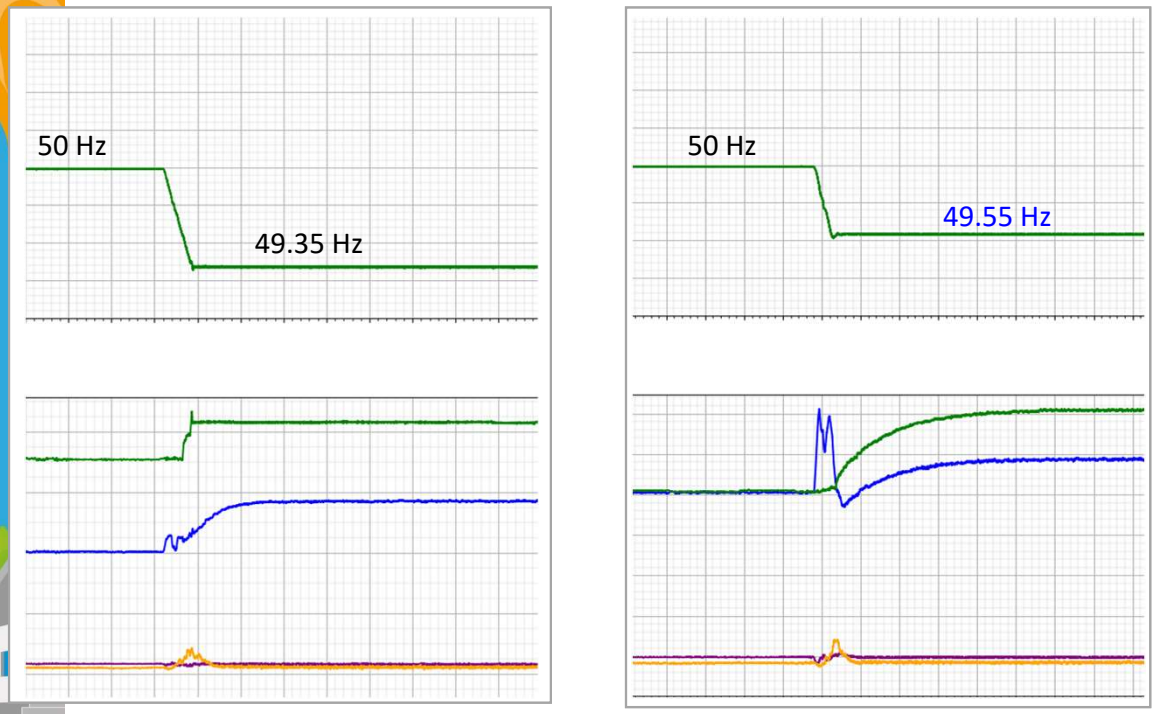


Synthetic inertia **Tj = 2 H = 12 s**

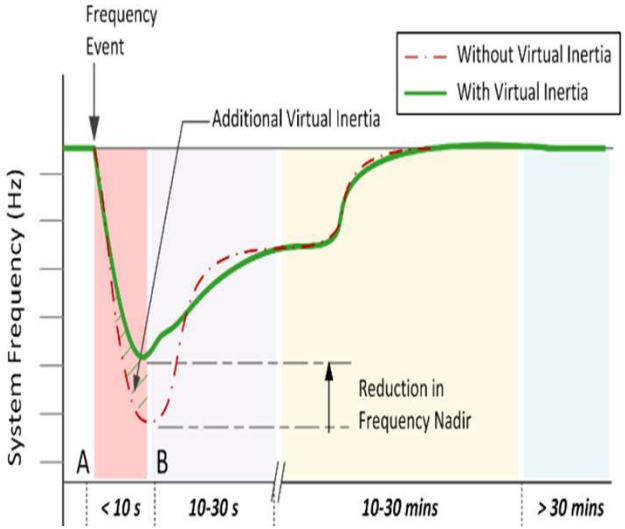
Frequency event: PCS in GFM mode + PV inverter + PPC

Low Virtual inertia H = 1.5

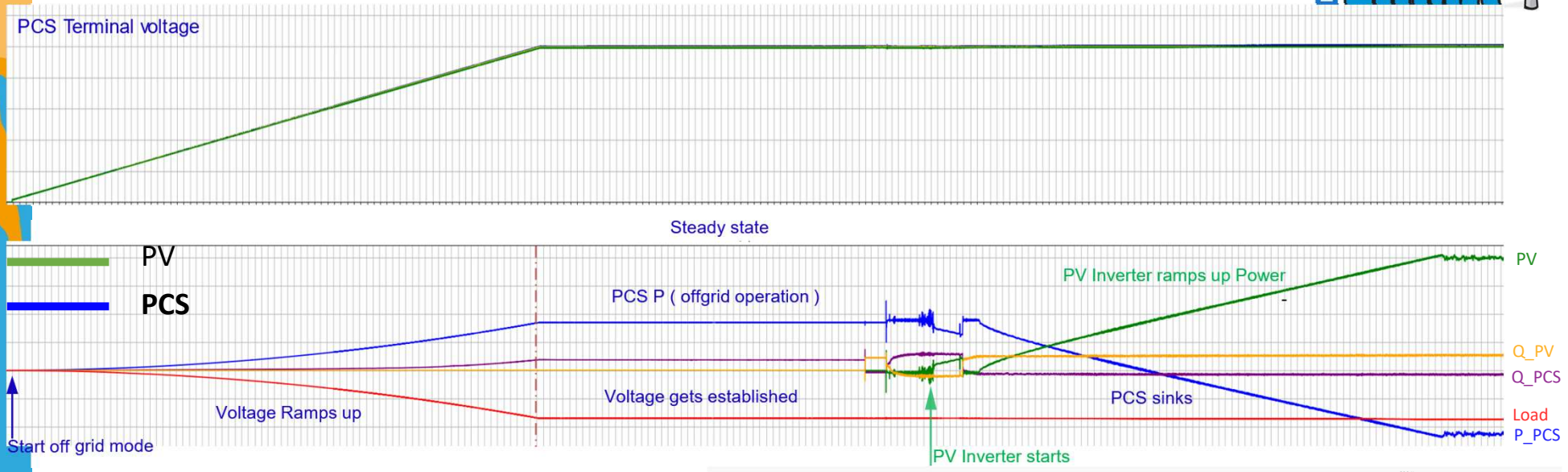
High Virtual inertia H = 6.5



Expected response in the grid

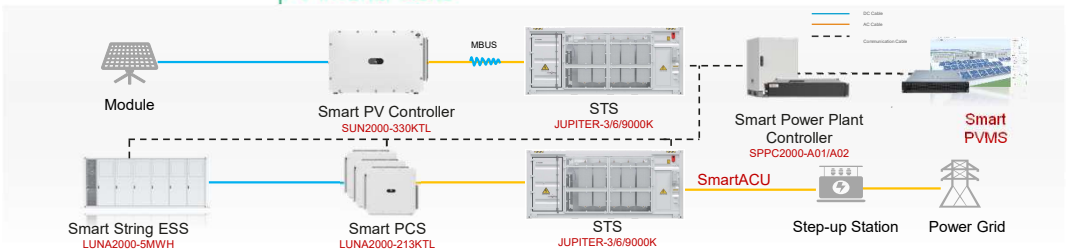


Black Start in a hybrid system: PV + PCS VSG + ESS system



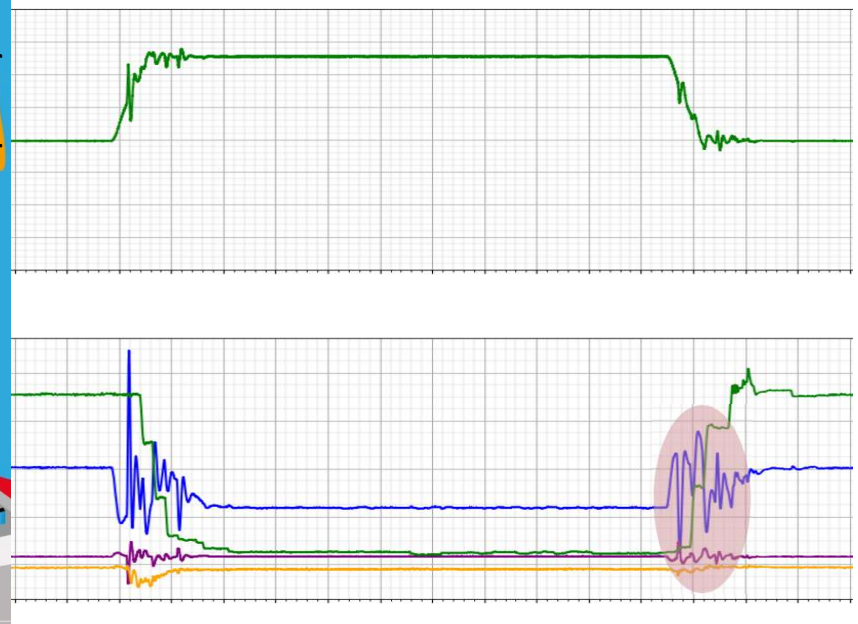
The **Network Code on Requirements for Grid Connection of Generators (NC RfG, Regulation (EU) 2016/631)** requires that Type C generating modules comply with:

- Article 15(5)(a) — generators must, if required by the TSO, **submit an offer to provide Black Start capability**
- It is **not mandatory by default** for all generating units, but **Member States may make this capability compulsory** if they consider it necessary for **system security**.



Stable coordination between the local fast dynamics of GFM converters and the higher-level control functions of the PPC.

Inertia response (PCS) + Primary freq control (PPC)



The PPC sent commands without accounting for GFM behavior, which destabilize the system.

Coordination GFM converter + PPC

ENTSO-E promotes a three-level control hierarchy:

- **Level 1**: GFM converter control (fast local voltage/frequency regulation)
- **Level 2:** Power Plant Controller (aggregated plant-level control)
- **Level 3:** System operator or market-level dispatch

This structure helps avoid controller conflicts and ensures stable operation even during faults or disturbances.

REE GFM proposal:

5.3 Casos de prueba y descripción de los eventos para evaluar la conformidad

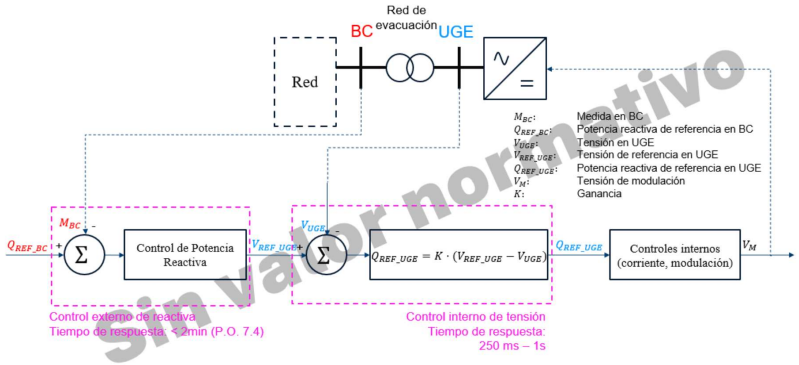
A continuación, se proponen las siguientes cuatro pruebas para evaluar las capacidades GFM, las cuales se explican en detalle en los siguientes epígrafes y se describe en detalle los eventos para evaluar la conformidad.

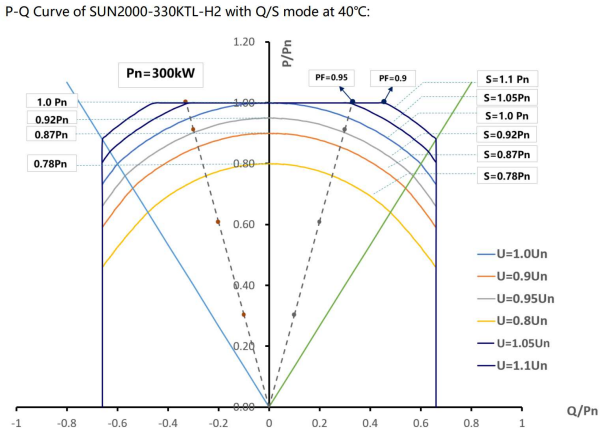
1. Comportamiento de la UGE como fuente de tensión dentro de los límites de capacidad.
2. Aporte de inercia sintética de la UGE dentro de sus límites de capacidad.
3. Comportamiento de la UGE cuando se alcanza el límite de capacidad.
4. Interacción entre controles de las UGE GFM.

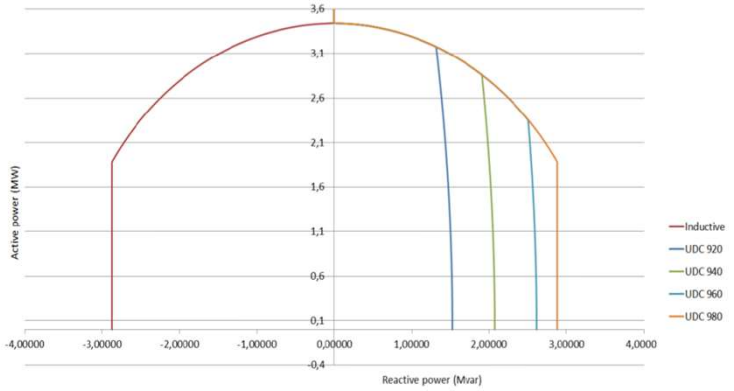
P.O.7.4 Servicio de control de tensión

En la actual propuesta de modificación del P.O. 12.2 se menciona lo siguiente respecto al modo de control combinado de potencia reactiva:

"Los módulos de parque eléctrico de tipo C y tipo D deberán disponer de un esquema combinado de control a consigna de potencia reactiva y a consigna de tensión"







GFM Standardization

Grid Code requirement and simulation models	•ENTSO-E guidelines inclusion in **Local Grid Codes** • Standard about dynamics simulation models •New ancillary service market
Core Dynamic control parameter for GFM	•Virtual inertia (**H**) and damping (**D**) •P–f / Q–V droop coordination •Voltage & frequency response time •Current limitation strategies
System Level coordination	•Virtual inertia interaction limits •Small signal and transient stability between control areas •Communication and coordination between the PPC and individual GFM units
Consistent grid studies	•Reproducibility between platforms •Validation with real data •Open real grid model for analysis and development

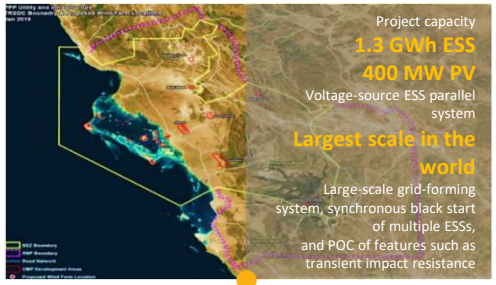
A harmonized GFM framework that supports **grid stability, certification consistency, and market integration**.

Since 2022, Huawei has completed multiple rounds of grid-forming ESS empirical tests

CR Power project in Gonghe, Qinghai
(World's first grid-forming PV+ESS test)

September 2022
First grid-forming PV+ESS test

The Red Sea destination in Saudi Arabia
(World's largest off-grid PV+ESS plant)



September 2023
100% renewable microgrid

CGDG project in Golmud, Qinghai
(World's first manual short-circuit test for the string grid-forming ESS plant)

June 2024
DC transmitter near-zone multi-energy complementary scenario

Complex and harsh environments

- Desert
- Cold
- High humidity
- Heat
- Salty
- High-altitude

Complex grid scenarios

- On-grid / offgrid
- Extremely weak grid
- Wind/Solar/BESS
- DC transmission terminal

May 2024
Wind power+ESS scenario

CR Power project in Hami
(World's first 100 MWh wind power+string grid-forming ESS plant performance test)

Jul, 2024
Extremely Weak Grid Terminal

ZDI project in Ngari Prefecture
(World's first manual short circuit test on high-altitude string grid-forming ESS)

Oct 2025
PV + ESS scenario

ENEL Matorral in Spain, Canary Island
(Spain First string grid-forming ESS in Utility)

Huawei's grid forming technology: Building six core capabilities for power generation, transmission, distribution, and consumption to redefine grid forming

Power generation side

Short-circuit current support

1–6x* vs. 1.2x
Reactive current

Inertia support

0–20s vs. 3–12s
Inertia time constant

Power transmission side

Wideband oscillation damping

0.1–100 Hz vs. 0.1–2.5 Hz

Primary frequency regulation

200 ms vs. 700 ms
Plant-level response time

Power usage side

Black start*

10 min@GWh vs. 1 h@MWh

On/Off-grid switching*

0 ms @ VSG
20 ms @ PQ to VSG

** PCS oversizing is required to achieve more than 1.25x short-circuit current capacity.*

** Black start and on/off-grid switching capabilities apply to microgrids.*

All plants

Plant-level
1000+ devices connected in parallel
GW-level, 100 km span

All working conditions

SOC range: 0–100% for constant power
SCR range: 1–40
Voltage range: 0–130% (HVRT/LVRT)
Phase angle range: 0–90°

All time domains

24/7
Steady-state process adjustment
Transient-state process support

Full lifecycle

One Matches All
Hardware capacity expansion and software iteration supported
A single platform designed to support future business models

HUAWEI

Huawei grid forming technology: Proven to be world-leading after rigorous testing

Smart grid forming control algorithm

String dual-stage architecture

Obvious advantages in key technical specifications

	Key indicators								
Huawei Grid Forming ESS Solution	Supported	SCR ≥ 1.0	Supported	Supported	Black start with 70% load	Multi-device 0.1–100 Hz	Supported	> 99.9%	Supported
	Plant-level short circuit: 1–6x overcurrent for 10s	Adaptable to the weak grid (field test)	Stable operation of over 1000 PCSs for grid forming	Transformer energization with 100% grid forming power capacity	Multi-device parallel system synchronous black start with loads	Field test of wideband oscillation damping in multi-device parallel connection	Constant active power during HVRT regardless of the SOC	ESS availability	Output at constant power regardless of the SOC
Other grid forming ESS solutions	/	/	/	/	No-load black start with a transformer	Only 0.1–2.5 Hz No multi-device field test reports	Not supported	< 99.0%	Not supported

Passed **more than 2300 plant-level tests**, including challenging short-circuit tests

World's first 100 MWh grid forming string ESS
Plant-level short-circuit disturbance test

Grid forming ESS for an extremely weak grid in Ngari Prefecture, Southwest China
Plant-level short-circuit disturbance test

Technical appraisal

Appraisal comment
The solution has achieved **world-leading capabilities** in terms of improving new power system stability and renewable energy integration and has achieved engineering application.

HUAWEI

1. Huawei Digital Power: Updates & Trends
2. Digitalization as an improvement of LCOS
3. Grid forming
4. **Intelligent O&M**

Current Situation of ESS Industry: Plant safety accidents occur frequently, and the safety prevention and control system is not perfect

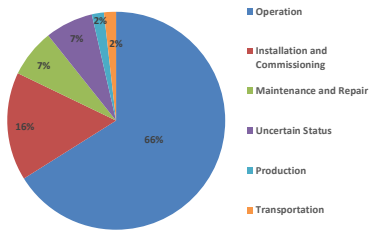
UK 165MWh ESS plant
Battery cell thermal runaway, direct loss of 2XX million yuan

Australia150MWh ESS plant
Environmental high-temperature thermal runaway, direct loss of $5XX million

China 200MWh ESS plant
Battery tip discharge, causing arc ignition and fire, direct loss of 5X million yuan, 1 person injured

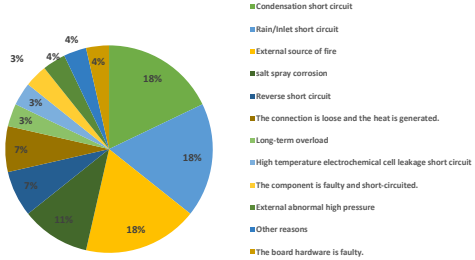
Moss Landing1200MWh ESS plant
Fourth fire incident in February 2025, resulting in over 70% equipment damage

2017-2024 E2E Incident Statistics of ESS



☐ **Operational phase incidents accounted for 66%**

2017-2024 ESS Accident Factor Analysis Statistics



☐ **Environmental and operational factors account for over 70%**

☐ **The safety prevention and control system of ESS plant is not perfect**

- ESS components are numerous and prone to diverse failures, **making manual maintenance tasks both burdensome and inefficient**
- Offline detection, passive maintenance, and fixed inspection cycles make it difficult to identify **operational risks.**
- The existing BMS itself carries a risk of failure and is unable to effectively identify **system-level faults**.
- The existing monitoring systems often employ a **patchwork approach** and lack adaptive capabilities in adjusting battery safety threshold parameters.

By leveraging mechanism algorithms and big data models, we can fully mine the data of ESS to establish evaluation and early warning mechanisms. Digital operation and maintenance methods play a significant role in reducing the risk of safety incidents.

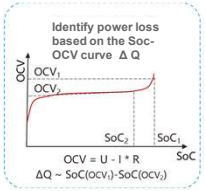
Comprehensive and professional ESS warning and diagnostic methods can help enhance product availability, delivering both safety and economic benefits.

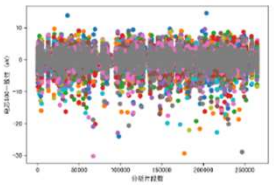
ESS faults can be detected in advance for proactive maintenance

Category	Exceptions can be intercepted	Typical problems can be intercepted
CELL	High cell voltage difference	The cell impurities cause self-discharge and generate low voltage, and the sampling circuit is damped, resulting in abnormal sampling
	High cell temperature difference	Abnormal NTC impedance/Soldering problem
	High cell SOC deviation	Clustered control SOC jumps due to accumulated integral errors of the SOC
	……	……
PACK	High pack temperature difference	Abnormal NTC impedance, EMS factory intelligent electric screwdriver fastening problem
	The fan is abnormal.	There are obvious friction marks on the fan blade steel sheet
	Abnormal insulation resistance	Cable harness is broken
	High pack SOH deviation	The voltage sampling deviation is continuously large
	……	……
ESS	Large temperature deviation of the pack inside the container	The return air temperature of the air conditioner in the control compartment is high, and the air conditioner is abnormal. The PACK reports a battery module temperature imbalance. The temperature of electrochemical cell 10 is lower than that of other electrochemical cells.
	Air conditioner fan speed detection	There is too much sand on the fan blade, which causes the fan to fail to run normally
	Compressor fault	Compressor drive fault, sensor fault, etc.
	Slight leakage of the liquid cooling pipe	Valve fault
	……	……

Preventive Alerting, Intelligent Operations

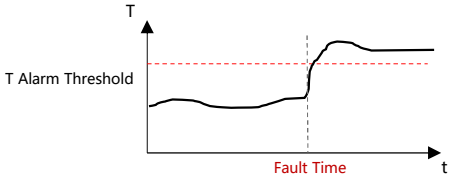
1. Complex mechanism algorithms and big data analysis to identify safety hazards in advance.



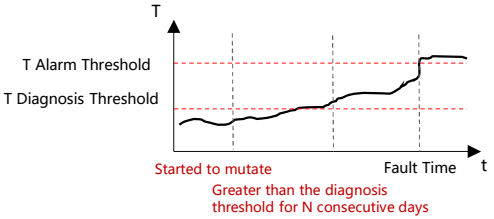


- Detects micro abnormal changes through **mechanism algorithms** and **data training**, and identifies potential risks in advance.

2. Identify issues at the board end in advance to avoid passive fault shutdowns.



Board-end alarm: sudden change of component stress triggers shutdown protection.



Diagnosis and warning: Slow variation of monitoring components or unhealthy O&M

3. Provide systematic and intelligent O&M service flows to improve O&M efficiency

Unification Diagnosis platform	**Comprehensive** Diagnosis Item	**Precise** Locking issues	**speciality** Handling suggestion

Smart risk warning and proactive protection based on full-link data collection, building a comprehensive ESS security protection network

Health Scoring and O&M Guide

Data processing, Diagnosis and warning

Full-link data collection

Higher availability

Warning 1-7 days in advance

Risk prediction and maintenance at a proper time, preventing unexpected shutdown from affecting production

Active safety

30+ risk identification

Real-time analysis and proactive protection, ensuring safe ESS operation

Assisted O&M

24h Real-time report

Detailed diagnosis report, providing guidance for ESS predictive maintenance

Long-period, refined data monitoring and indicator-based warning

- ✓ The device undergoes slow variations, and damage will lead to equipment shutdown. By **analyzing data trends**, anomalies can be identified in advance.
- ✓ It **covers various operating conditions** such as peak shaving, frequency regulation, microgrids, and backup power, providing real-time monitoring and assessing the health of the system.
- ✓ **Long-period data statistics and analysis ,** support O&M personnel in conducting planned proactive O&M, ensuring long-term safe and high-quality operation.

Big data model training

battery mechanism model

multi-physical field coupling model

Battery big data model

- ✓ Collect long-period, fine-grained full-link data to support algorithm and model training, deploy independent servers locally, and provide a **computing platform capable of handling tens of millions of data points**.
- ✓ Microscopic abnormal changes in components cannot be detected through conventional inspection and maintenance. Integrate **over 10 mechanistic algorithms and big data models** for monitoring and judgment.
- ✓ 100% self-developed, with **23** algorithm patents.

8000+次/天

Cell	Pack	Rack	LTMS	ESS	PCS	Array	Grid Connection Point*
Cell current monitoring Cell temperature monitoring Cell voltage deviation analysis Cell SOH deviation analysis Cell SOC deviation analysis Cell internal and external short circuit warning	Pack current monitoring Pack temperature monitoring Pack voltage deviation analysis Pack SOH deviation analysis Pack SOC deviation analysis	Rack current monitoring Rack voltage monitoring Charge/Discharge power curve analysis EEV action monitoring	LTMS Compressor status LTMS Refrigerant temperature	ESS internal temperature monitoring ESS internal humidity monitoring EEV action monitoring ESS derating analysis ESS insulation resistance monitoring SOH coefficient of variation analysis Array insulation resistance monitoring	PCS internal temperature monitoring PCS internal humidity monitoring PCS Insulation resistance monitoring	Array Insulation resistance monitoring	Grid Connection Point Voltage Monitoring Grid Connection Point Current Monitoring Fault Recording...

*The monitoring at the grid connection point requires the supplementary capabilities of SPPC device.

Smart Health Guard Working Process

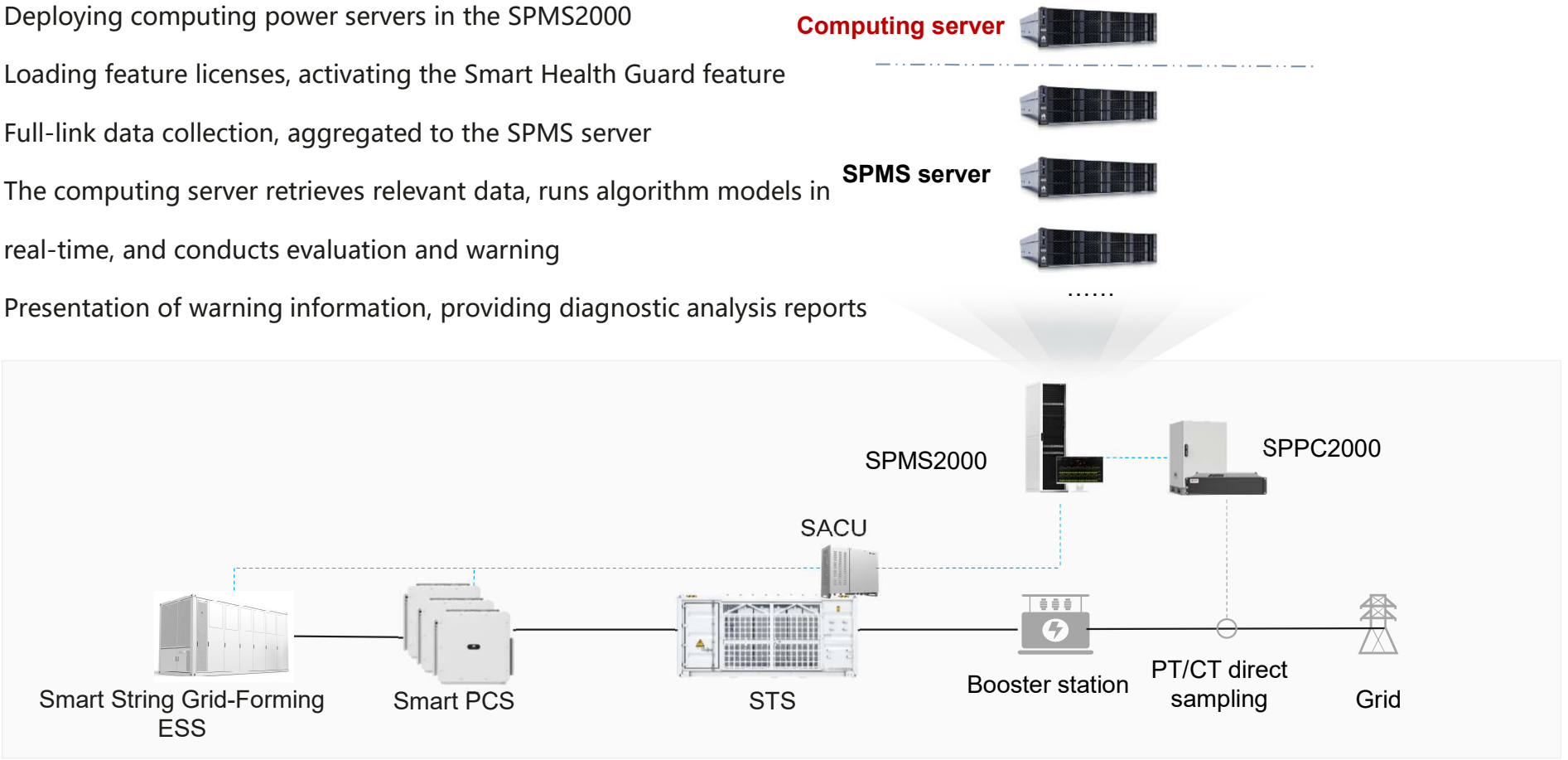
① Deploying computing power servers in the SPMS2000

② Loading feature licenses, activating the Smart Health Guard feature

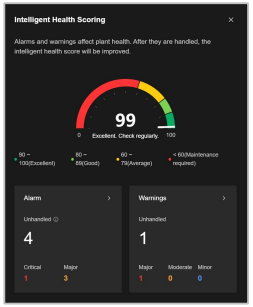
③ Full-link data collection, aggregated to the SPMS server

④ The computing server retrieves relevant data, runs algorithm models in real-time, and conducts evaluation and warning

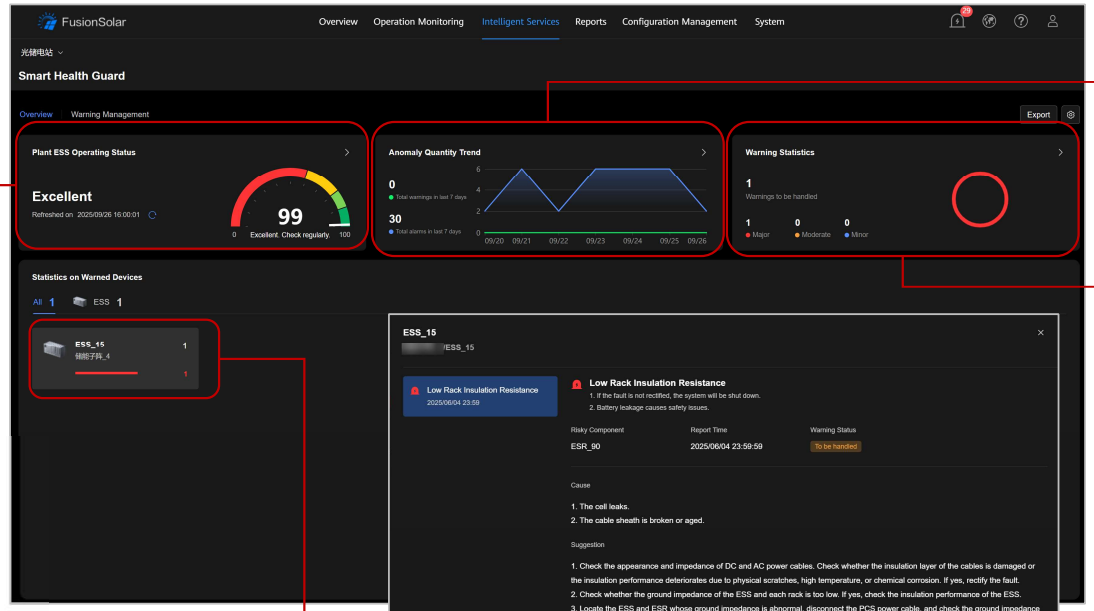
⑤ Presentation of warning information, providing diagnostic analysis reports

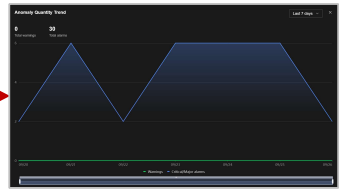


Smart Health Guard，Scoring the health check of the plant to ensure the safe operation of the ESS

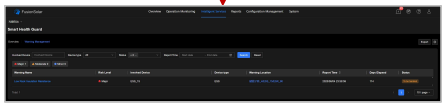


Continuously monitor the plant status and score its health condition.





Total warning statistics, alert statistics, trend analysis of warning quantities, warning classification



Warning management, view abnormal device distribution, support inspection plan customization

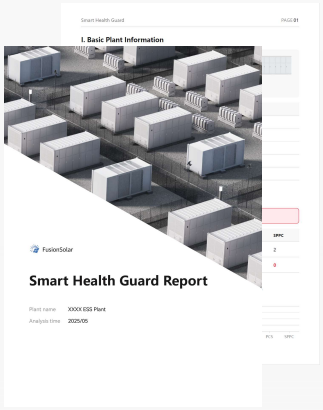
SPMS2000

Detailed warning Analysis

- Warning device information to accurately locate faults
- Exception Cause Analysis and Handling Suggestions
- Abnormal Data and Mark Analysis

Diagnosis reports are generated in one-click mode to guide predictive maintenance

Detailed diagnosis report

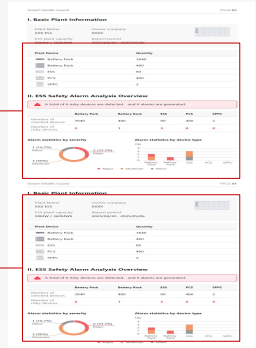


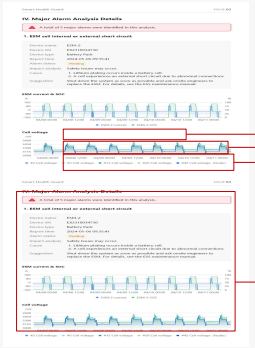
Quarterly, monthly, and daily dimensions

Smart Health Guard Report One-click export

Basic Plant Information

Risk Overview, Early Warning Statistics





Impact Analysis

Root Cause Identification

O&M Suggestions

Data curve

Real-time analysis and proactive protection

Identifies 30+ security risks.

Ensure the safe running of the entire ESS

No.	Fault Type	Object	No.	Fault Type	Object
1	Cell internal and external short circuits	Cell	9	Abnormal temperature sensor sampling	LTMS
2	Cell high/low voltage and high/low temperature	Cell	10	Pressure sensor sampling abnormality	LTMS
3	Abnormal cell SOC	Cell	11	ESS internal abnormal temperature	ESS
5	High cell voltage and temperature difference in ESM	ESM	12	ESS internal abnorma humidity	ESS
6	High ESM cell SOH deviation	ESM	13	Not charged or discharged for a long time	ESS
7	High ESM cell SOH deviation	ESM	14	Alarm O&M reminder	ESS array
8	High ESM SOC deviation in the ESR	ESR		...	

Online monitoring and Offline inspection ensure the long-term stability of ESS plant

Smart Health Guard Risk Overview

- Abnormal Device Distribution
- Abnormal Type Statistics
- Trend Analysis
- ...

Inspection Preparation& solution formulation

- Inspection Schedule Customization
- Downtime Plan
- Spare Parts and Personnel Preparation
- ...

Key Equipment and Operating Status Inspection

- Health Guard provides risk-related O&M recommendations.
- .....

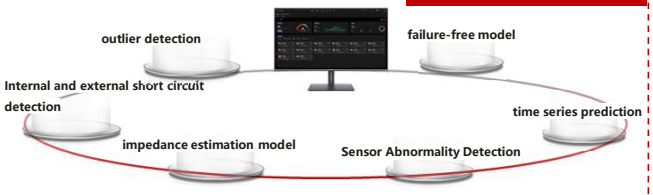
Software Logs Inspection

- Health Guard provides risk-related O&M recommendations.
- .....

Output the professional inspection report

Based on the diagnostic results and risk warning distribution obtained through online monitoring, a customized offline expert inspection plan is developed to conduct more thorough and comprehensive inspections of ESS plant. Comprehensive warning reports and inspection results are provided to offer an all-round, full-lifecycle guarantee for the safe, stable, and healthy operation of ESS device

Smart Health Guard



Online monitoring, risk warning, intelligent analysis, online viewing, report generation

Discover potential risks in plant operation in advance and handle risks in a timely manner and provide guidance for customers to perform proactive O&M.

+

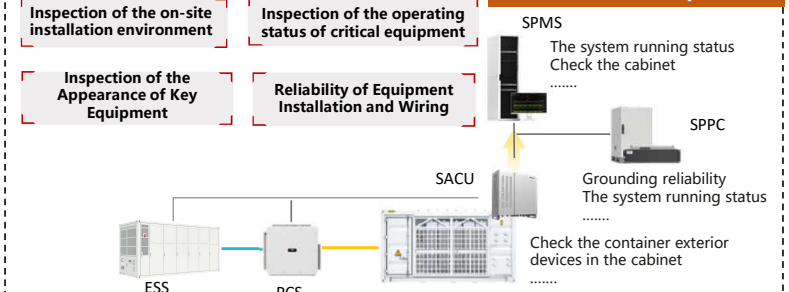
Offline device inspection

Inspection of the on-site installation environment

Inspection of the operating status of critical equipment

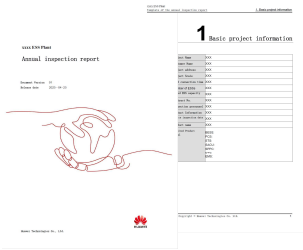
Inspection of the Appearance of Key Equipment

Reliability of Equipment Installation and Wiring



Professional on-site inspections, timely handling of safety hazards, and providing professional rectification and maintenance recommendations.

Final Inspection Report



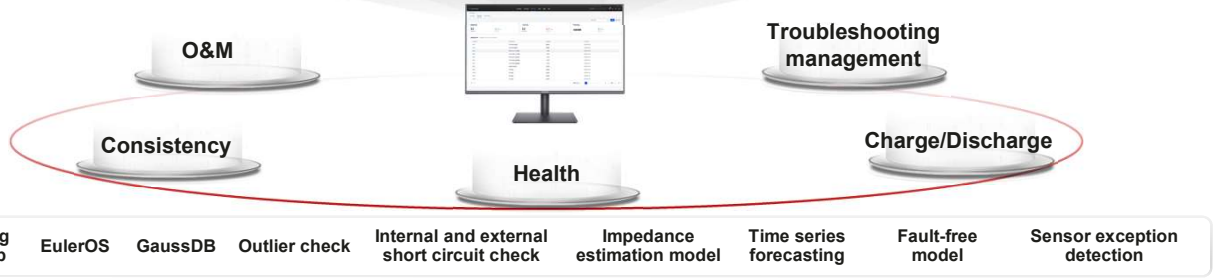
Online and offline collaborative analysis, providing professional inspection reports

ESS health safety guard: AI-based risk warning and proactive protection, ensuring safety

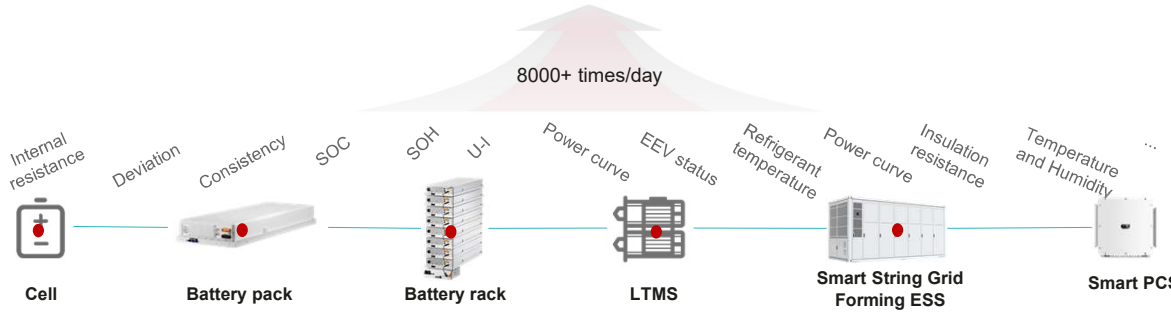
Three key features

Higher availability	Active safety	Auxiliary O&M
Early warning (1–7 days in advance): Risk prediction for targeted maintenance, preventing unexpected shutdown that may affect production	**Identification of 30+ potential risks**: Real-time analysis and proactive protection, ensuring safe ESS operation	Detailed diagnosis report, providing guidance for ESS predictive maintenance

Active safety algorithm platform



Full-link data collection



** To be launched in H2 2025, not for sales now*

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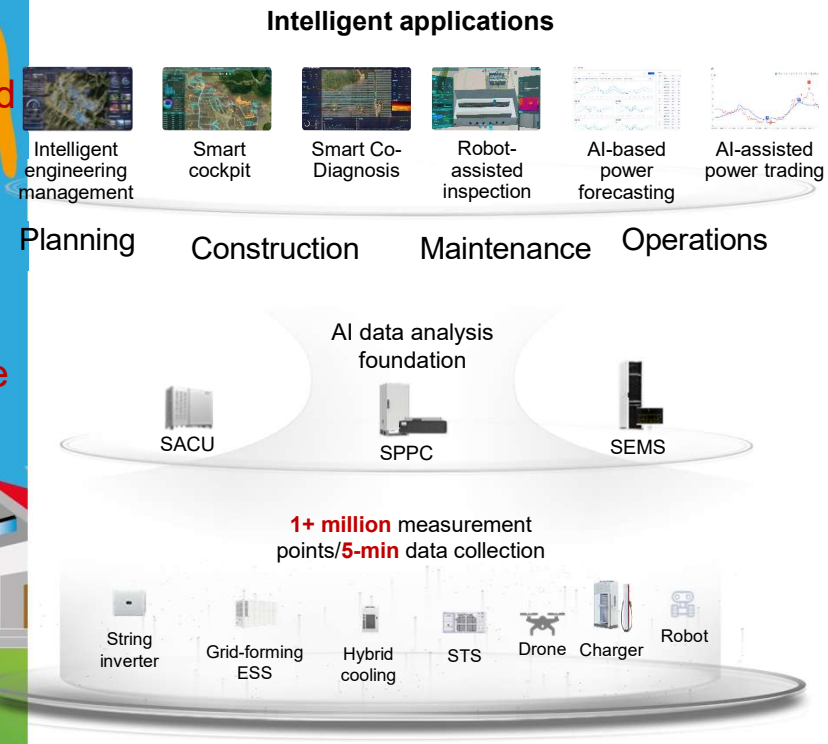
FusionSolar Agent: End-to-end intelligent plant lifecycle management powered by AI

Visible Overview of Intelligent O&M and Grid Forming

AI is penetrating into the entire link and lifecycle of plants



Shache Energy Storage Plant: Smart health guard, safe and long-lasting

Around-the-clock online diagnosis | 7-day warning | C2G health evaluation

Benefits

- > 90% Accuracy and recall rate
- 100 million+ times per day AI data processing
- 30% Higher O&M efficiency
- 20%+ ESS life extension

Visible overview of grid forming capabilities (New)

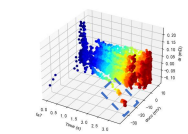
- Comprehensive grid forming features of power plants, local scheduling of active and reactive power, and real-time response
- HVRT/LVRT, low-frequency oscillation, and renewable energy frequency regulation trigger level-3 automatic recording.
- Pay attention to the active power, reactive power, and power factor at the PCC.

* The level-3 fault recording capability depends on the connection of the SPPC/SmartLogger5000A/SUN2000HD.

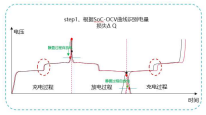
* The grid forming capability depends on the update of the SPPC and PCS software versions to PCS2000HA V200R024C10 or later.

HUAWEI

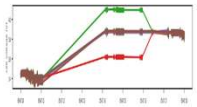
A warning is generated 1 to 7 days before a cell problem occurs, with an accuracy of > 90%*



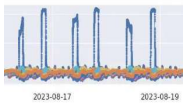
Abnormal SOC and impedance



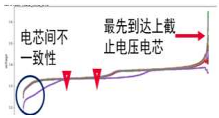
Internal/External short circuit of the battery



Abnormal sensor sampling



Battery abuse



Abnormal SOH

Huawei 2012 Labs' intelligent algorithm expert team has years of algorithm experience.

SPMS - Data processing

Powerful computing capability
Processing tens of millions of cell data records

Big data
Data governance + feature engineering

Many models
Mechanism and multi-physical field model

Device management and system collaboration, no need to migrate data to the cloud

Device - data collection

High accuracy
±1.5 mV sampling precision
Devices delivered after September 30, 2025 meet the requirements.

High frequency
10s sampling frequency

Small granularity
Battery data sampling

1 200 MWh ESS project in Hubei

- Running time > 12 months
- Dozens of high-risk exceptions have been warned, such as SOC consistency exception, voltage sampling exception, and temperature sampling exception.

2 50 MWh ESS project in Hainan

- Running time > 4 months
- Four high-risk problems have been warned, such as SOC consistency exception.

3 8 MWh ESS project in Guangdong

- Running time > 9 months
- Nine warnings have been generated for internal and external short-circuit risks and SOC consistency exceptions, among which one is a high risk.

4 300 MWh ESS project in Uzbekistan

- Running time > 3 months
- Two warnings have been generated for SOC consistency exception.

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Reducing the number of onsite visits, simplifying operations, and improving efficiency by more than 50%

Free onsite check and update services

Convenient and traceable online management of major issues

[Black box]

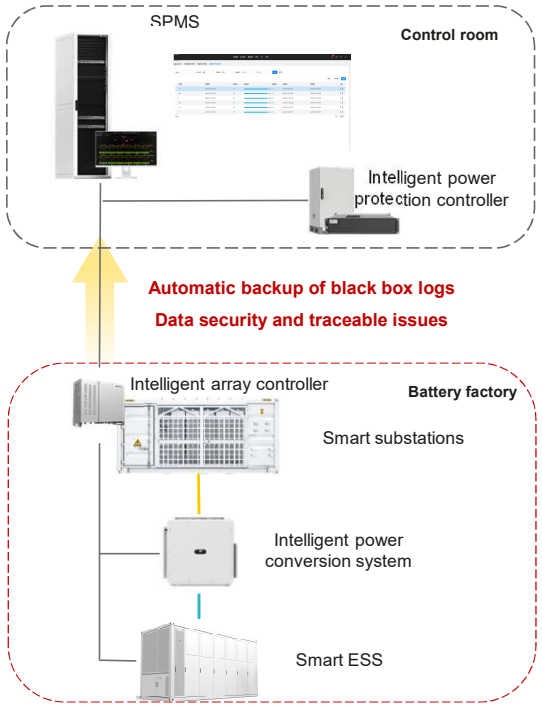
- Key logs such as overtemperature and fan faults are stored in the EMS.
- When a major fault occurs, data can be exported to locate and analyze the fault cause.

[Remote device inspection]

- Online inspection on current, voltage, insulation resistance, and internal temperature

[Remote device management]

- Online log export
- Online device batch upgrade



Convenient troubleshooting and resolution

Long-term alarm record storage facilitates O&M.

[Historical alarms]

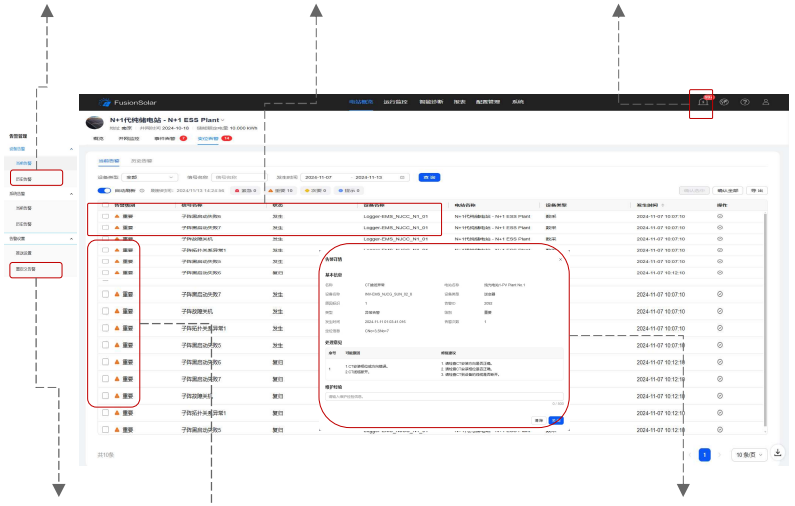
- Historical alarm storage period > 2 years

[Alarm filtering]

- Quick alarm search

[Alarm shortcut]

- One-click access to the alarm center



[Redefining]

- Customized event alarm priority

[Severity]

- Major alarms displayed first

[Alarm details]

- Detailed handling suggestions
- Record of handling experience

Thank you